

The logo for Fiveable, featuring the word "fiveable" in a bold, sans-serif font. The word is split into "five" in white and "able" in dark blue, set against a light blue speech bubble background.

fiveable

AP Physics 1

Finale Part 1--Energy!

Head's up!

This stream is being recorded and might appear on our website, YouTube, or other public channels.

If you ask a question, your name will appear on the screen or be read.
Please change your name if you want to avoid this 👍

@thinkfiveable



- **5:00-5:55 Energy!**
- **6:00-6:55 Momentum!**
- **7:00-7:55 Simple Harmonic Motion**
- **8:00-8:55 FRQ Tips and Tricks**
- **9:00-9:00 MCQ Tips and Tricks
with GimKit Challenge!**

Section I: Multiple Choice

50 Questions | 1 Hour 30 Minutes | 50% of Exam Score

- Questions are either discrete questions or question sets, in which students are provided with a stimulus or a set of data and a series of related questions.
- The section includes 5 individual multi-select questions (2 options are correct).

Section II: Free Response

5 Questions | 1 Hour 30 Minutes | 50% of Exam Score

- This section contains 5 free-response questions of the following types:
 - Experimental Design (1 question)
 - Qualitative/Quantitative Translation (1 question)
 - Short Answer: Paragraph Argument (1 question)
 - Short Answer (2 questions)

Things you NEED:

1. Calculator! (just can't have a qwerty keyboard)
 - a. Graphing calculators are FINE!
2. PENCIL...a REGULAR Pencil (NO MECHANICAL)
3. A RULER (YES! You MAY use one)...and USE THE RULER!
 - a. Free-body diagrams
 - b. Best-fit lines on graphs

The Equation Sheet

Cross off questions that you will NEVER use on this exam!

MECHANICS

$$v_x = v_{x0} + a_x t$$

a = acceleration

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

A = amplitude

d = distance

E = energy

f = frequency

F = force

I = rotational inertia

K = kinetic energy

k = spring constant

L = angular momentum

ℓ = length

m = mass

P = power

p = momentum

r = radius or separation

T = period

t = time

U = potential energy

V = volume

v = speed

W = work done on a system

x = position

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_n|$$

$$a_c = \frac{v^2}{r}$$

$$\vec{p} = m\vec{v}$$

$$\Delta \vec{p} = \vec{F} \Delta t$$

$$K = \frac{1}{2} mv^2$$

$$\Delta E = W = F_{\parallel} d = F d \cos \theta$$

$$\Delta E$$

ELECTRICITY

$$|\vec{F}_E| = k \left| \frac{q_1 q_2}{r^2} \right|$$

A = area

F = force

I = current

ℓ = length

P = power

q = charge

R = resistance

r = separation

t = time

V = electric potential

ρ = resistivity

$$I = \frac{\Delta q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I \Delta V$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

WAVES

$$\lambda = \frac{v}{f}$$

f = frequency

v = speed

λ = wavelength



Students, draw anywhere on this slide!

The Equation Sheet

Don't forget the
math based
help section!

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

Rectangular solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

A = area

C = circumference

V = volume

S = surface area

b = base

h = height

ℓ = length

w = width

r = radius

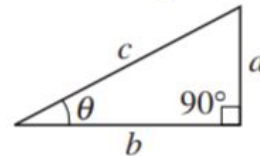
Right triangle

$$c^2 = a^2 + b^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$



The Equation Sheet

Constants are given--you MAY use 10 m/s^2 for acceleration due to gravity! *Cross off items you don't need for this exam!*

ADVANCED PLACEMENT PHYSICS 1 EQUATIONS, EFFECTIVE 2015

CONSTANTS AND CONVERSION FACTORS

Proton mass, $m_p = 1.67 \times 10^{-27} \text{ kg}$	Electron charge magnitude, $e = 1.60 \times 10^{-19} \text{ C}$
Neutron mass, $m_n = 1.67 \times 10^{-27} \text{ kg}$	Coulomb's law constant, $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
Electron mass, $m_e = 9.11 \times 10^{-31} \text{ kg}$	Universal gravitational constant, $G = 6.67 \times 10^{-11} \text{ m}^3/\text{kg}\cdot\text{s}^2$
Speed of light, $c = 3.00 \times 10^8 \text{ m/s}$	Acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$

UNIT SYMBOLS	meter, m	kelvin, K	watt, W	degree Celsius, °C
	kilogram, kg	hertz, Hz	coulomb, C	
	second, s	newton, N	volt, V	
	ampere, A	joule, J	ohm, Ω	



Students, draw anywhere on this slide!

The Equation Sheet

The prefixes and the CONVENTIONS are useful! Air resistance is NEGLIGIBLE!
Positive work is defined as work done **ON** a system.

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan \theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following conventions are used in this exam.

- I. The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- II. Assume air resistance is negligible unless otherwise stated.
- III. In all situations, positive work is defined as work done on a system.
- IV. The direction of current is conventional current: the direction in which positive charge would drift.
- V. Assume all batteries and meters are ideal unless otherwise stated.

Work, Energy, and POWER!



$$\Delta U_g = mg \Delta y \quad U_s = \frac{1}{2} kx^2 \quad K = \frac{1}{2} mv^2 \quad K = \frac{1}{2} I\omega^2$$

$$U_G = -\frac{Gm_1m_2}{r}$$

$$\Delta E = W = F_{\parallel} d = Fd \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

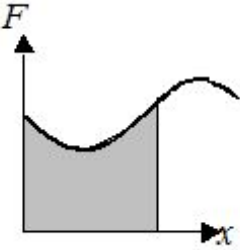
Important Quantities

Name	Symbol	Units	Basic Equation
Work	W	J	$W = Fx_{ }$
Power	P	W	$P = \frac{\text{Energy}}{\text{time}}$
Energy	E	J	None

Name	Symbol	Units	Basic Equation
Kinetic Energy	K	J	$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$
Potential Energy (Gravity)	U_g	J	$U_g = mgh$
Potential Energy (Spring)	U_s	J	$U_s = \frac{1}{2}kx^2$

Important Equations

Name	Equation	Given?	Notes
Work done by Friction (also energy lost to friction)	$W_f = F_f x$	No	Friction takes some energy out of the system. Note that energy lost to friction becomes heat, so sometimes this is referred to as <u>thermal</u> energy.
Conservation of Energy	$K_i + U_i = K_f + U_f$	No	The sum of all different energies before equals the sum of all different energies after. These could be kinetic, potential, thermal, etc. energies.
Power (in terms of Force and Velocity)	$P = Fv_{\parallel}$	Yes	Note that dot products imply that you are multiplying the two vectors and the “cosine of the angle between them”.

Name	Graph (Shape)	Notes
Force vs. Displacement	 (Could be anything)	The area under a force vs. displacement graph is work.

IMPORTANT CONCEPTS

- NO WORK IS DONE IF FORCE IS PERPENDICULAR TO AN OBJECT MOTION!!!
- USE ENERGY EQUATIONS IF YOU ARE ASKED FOR VELOCITY IN TERMS OF POSITION, OR POSITION IN TERMS OF VELOCITY!
- Note that the equations $W = Fx_{\parallel}$ and $P = Fv_{\parallel}$ are equivalent to $W = F_{\parallel}x$ and $P = F_{\parallel}v$
- and $W = Fx \cos \theta$ and $P = Fv \cos \theta$
 - Cosine does the job of finding the parallel component for you.

Important Concepts: + and - Work

- Work is a transfer of energy from one form to another. If energy goes from kinetic to potential, potential to kinetic, kinetic to thermal, kinetic to another object's kinetic, etc., work was done.
- Work is a quantity measured in joules (like energy) because the amount of work is the amount of energy that was transferred.
- If work is **positive**, then that is a **gain** in energy (getting more energy or an increase in energy).
- If work is **negative**, then that is a **loss** of energy (giving away energy or a decrease in energy).

Important Concepts: + and - Work

A force does work if the force acts on the object while the object moves from one point to another. The sign of work depends on the angle between the direction of the force and the direction of the displacement:

- Work is positive if the direction of force and direction of displacement are acute (less than 90°)
- Work is negative if the direction of force and direction of displacement are obtuse (more than 90° up to 180°)
- No work is done by a force if the direction of force and direction of displacement are **perpendicular**.

Important Concepts: + and - Work

- Potential energy increases if an object is forced to do “the opposite of what it wants to do.” Likewise, potential energy decreases if an object is allowed to do “what it wants to do.”
 - Objects want to fall down. So raising an object up increases its potential energy.
 - Springs don't like to be stretched or compressed. Therefore, compressing or stretching a spring (away from its equilibrium position) increases its potential energy.
 - Objects in space want to attract each other due to gravity. Therefore, moving objects apart in space increases their potential energy.

BE CAREFUL!

- The work done ON an object is POSITIVE if the potential energy increases.
- The work done BY the force is NEGATIVE if the potential energy increases.

Important Concepts: + and - Work

Example:

I lift a book from a low shelf to a high shelf.

- I do positive work (on the book) because the force I exert (up) is in the same direction as displacement (up)
- My potential energy decreases because I had to burn calories to lift the book but the potential energy of the book increases
- On the other hand, the force due to gravity did negative work because the force of gravity (down) was opposite the direction of displacement (up)
- Gravitational potential energy increased

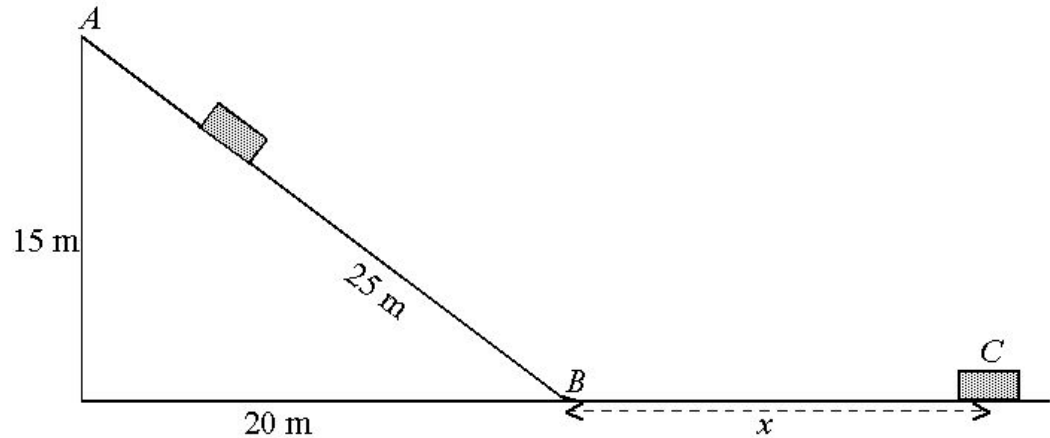
NO WORK IS DONE ON AN OBJECT IN ANY UNIFORM CIRCULAR MOTION (the speed doesn't change, so no work is done because no energy is transferred). THIS INCLUDES A PLANET IN CIRCULAR ORBIT.

Big concept: Use energy whenever are given a position and asked for a velocity, OR given a velocity and asked for a position.

Important Concepts: Incline Plane

Example: A block is released from rest at the top of the incline (point A) as shown.

A. If the incline is frictionless, how fast does the block move at point B?



Students, draw anywhere on this slide!

Important Concepts: Incline Plane

Example: A block is released from rest at the top of the incline (point A) as shown.

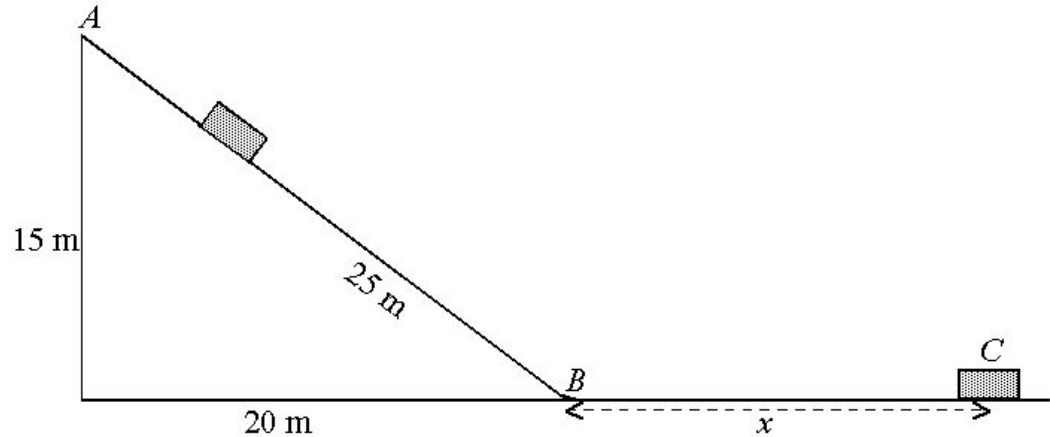
A. If the incline is frictionless, how fast does the block move at point B?

$$U_g = K$$

$$mgh = \frac{1}{2}mv^2$$

$$v = \sqrt{2gh} = \sqrt{2(10)(15)}$$

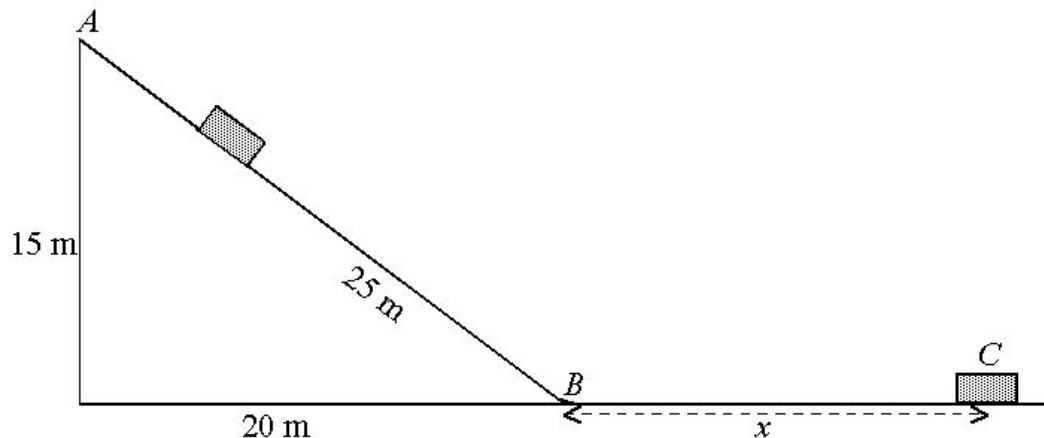
$$v = 17.3 \text{ m/s}$$



Important Concepts: Incline Plane

Example: A block is released from rest at the top of the incline (point A) as shown.

B. If the incline has a coefficient of friction of 0.11, what is the speed at point B?



Students, draw anywhere on this slide!

Important Concepts: Incline Plane

Example: A block is released from rest at the top of the incline (point A) as shown.

A. If the incline has a coefficient of friction of 0.11, what is the speed at point B?

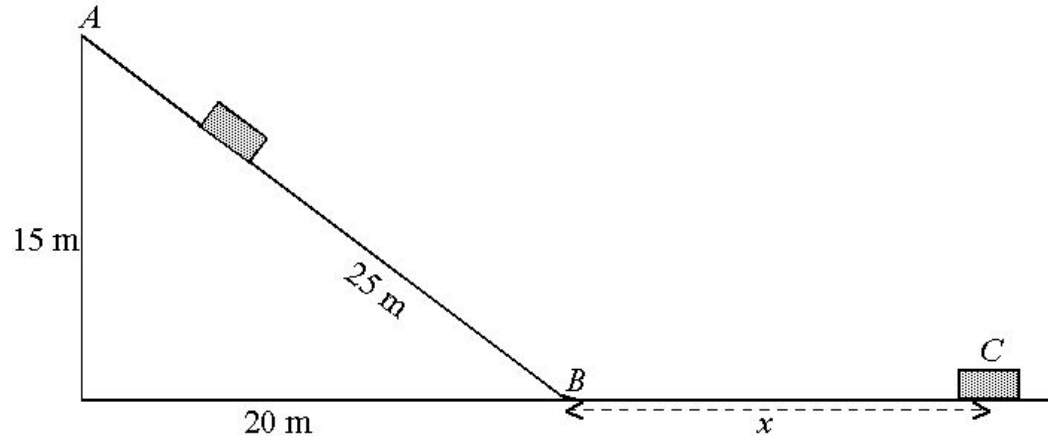
$$mgh = \frac{1}{2}mv^2 + F_f x$$

$$F_f = \mu F_N = \mu mg \cos \theta$$

$$mgh = \frac{1}{2}mv^2 + \mu mgx \cos \theta$$

$$v = \sqrt{2gh - 2\mu g x \cos \theta} = \sqrt{2(10)(15) - 2(0.11)(10)(25)\left(\frac{20}{25}\right)}$$

$$v = 16 \text{ m/s}$$

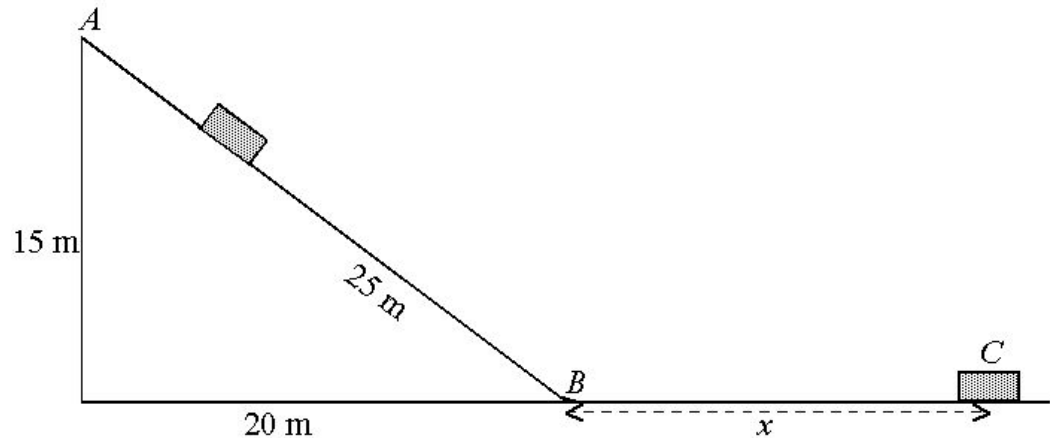


Important Concepts: Incline Plane

Example: A block is released from rest at the top of the incline (point A) as shown.

C. If the incline and the horizontal plane have a coefficient of friction of 0.11, what is the distance “x” traveled along the horizontal surface to point C?

Speed at B is 16 m/s



Students, draw anywhere on this slide!

Important Concepts: Incline Plane

Example: A block is released from rest at the top of the incline (point A) as shown.

C. If the incline and the horizontal plane have a coefficient of friction of 0.11, what is the distance “x” traveled along the horizontal surface to point C?

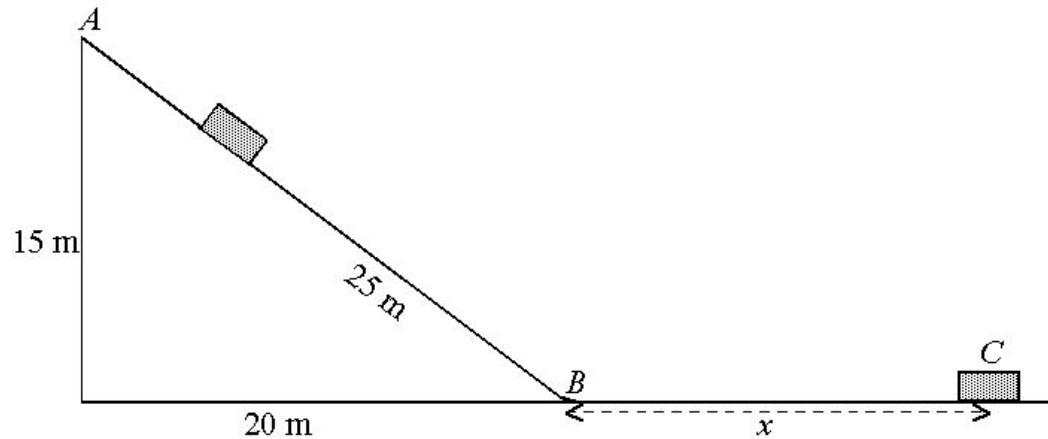
Speed at B is 16 m/s

$$K_B = W_{\text{friction}} \text{ at "C"}$$

$$\frac{1}{2} mv^2 = \mu mgx$$

$$\frac{1}{2}(16)^2 = (.11)(10)x$$

$$x = 116 \text{ m}$$

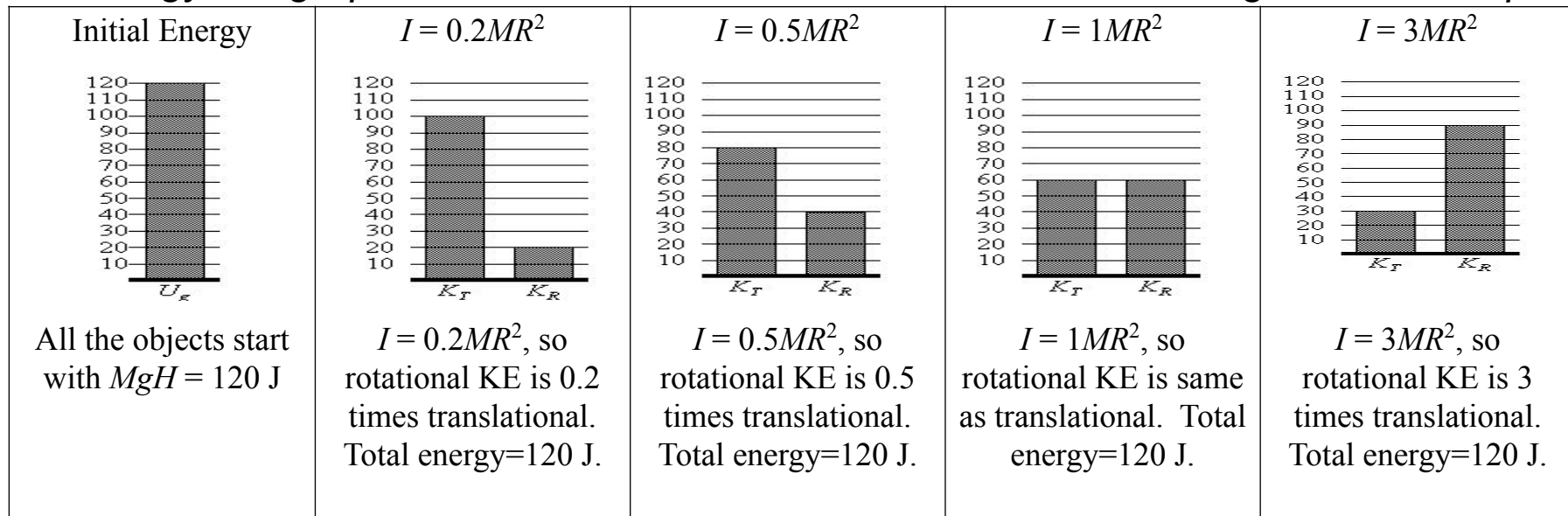


ROLLING OBJECTS:

- When an object rolls without slipping, it has both translational and rotational kinetic energy.
- If an object has rotational inertia kmr^2 , then the rotational kinetic energy is k times the translational kinetic energy.
 - For example, an object whose rotational inertia is $\frac{1}{4}mr^2$ has $K_R = \frac{1}{4}K_T$ at all times while it rolls without slipping.

Important Concepts: Rolling Objects

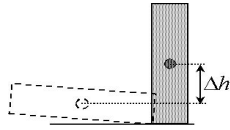
Consider four objects with the same mass and that start at rest at the top of the same ramp. Suppose that they start with 120 J of potential energy. Now the objects roll down the ramp to the bottom. Here is what each object's final energy bar graph looks like. Note that their rotational inertia is given at the top.



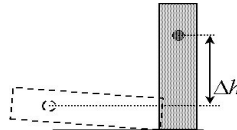
Important Concepts: Potential Energy from Center of Mass location

POTENTIAL ENERGY IS ALWAYS MEASURED FROM THE OBJECT'S CENTER OF MASS.

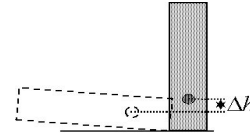
If a bookcase falls over (starts with gravitational potential energy and ends with rotational kinetic energy)



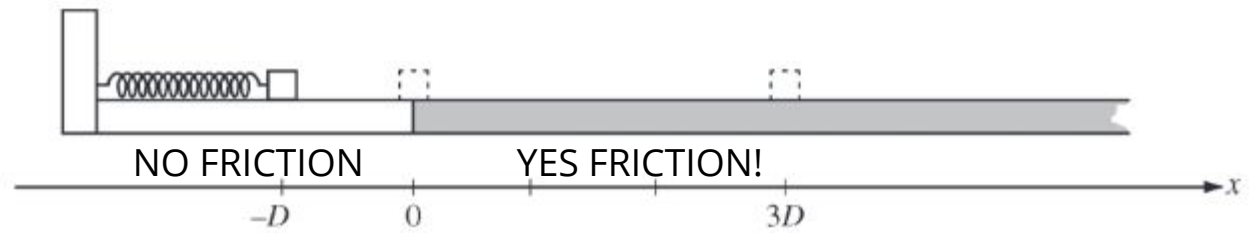
A bookcase without books has a center of mass in the center of the bookcase. The bookcase's mgh is found by measuring the height of the center of mass.



Suppose there are books on the top shelf. These books put the center of mass high, so the bookcase has more potential energy to start with and more KE when it falls over.

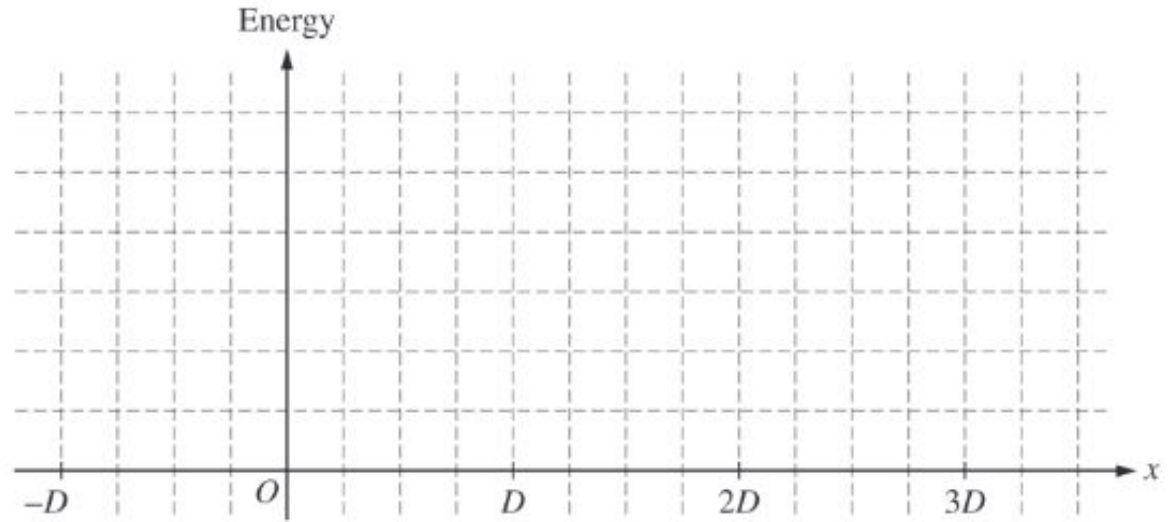


Suppose there are books on the bottom shelf. These books put the center of mass low, so the bookcase has less potential energy to start with and less KE when it falls over.

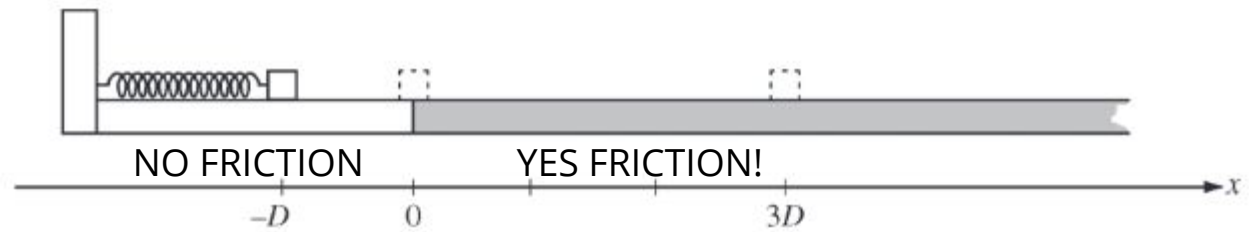


(a) On the axes below, sketch and label graphs of the following two quantities as a function of the position of the block between $x = -D$ and $x = 3D$. You do not need to calculate values for the vertical axis, but the same vertical scale should be used for both quantities.

- The kinetic energy K of the block
- The potential energy U of the block-spring system

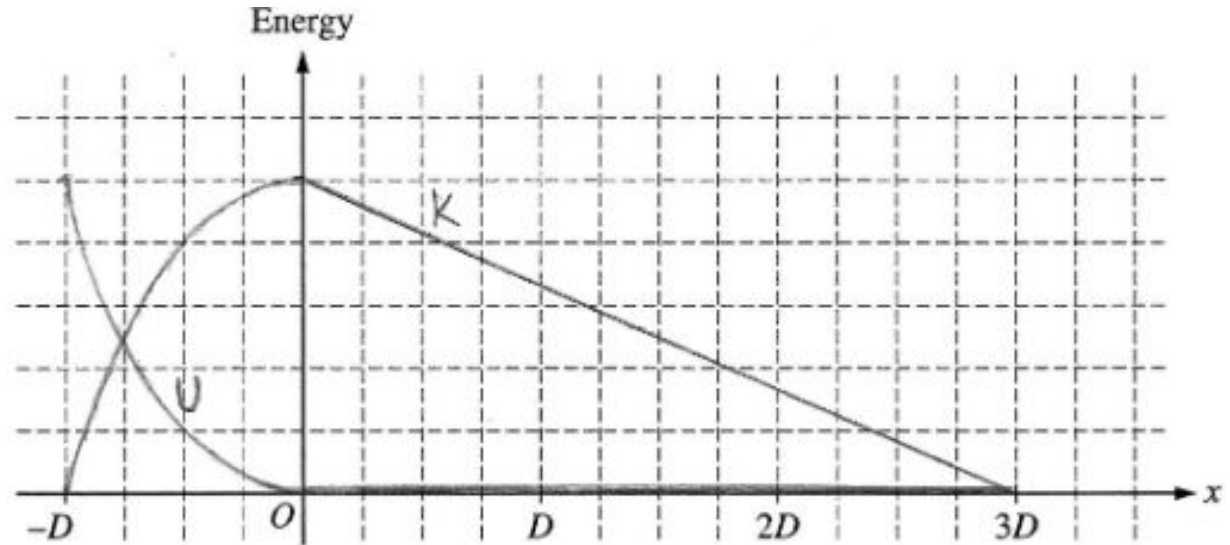


Students, draw anywhere on this slide!



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- The kinetic energy K of the block
- The potential energy U of the block-spring system



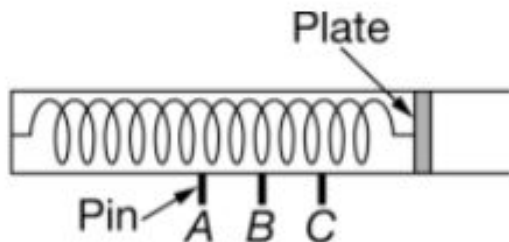


Figure 1. Uncompressed spring

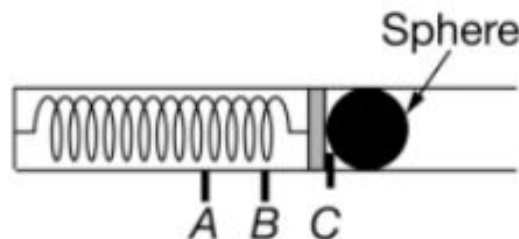


Figure 2. Compressed spring

(12 points, suggested time 25 minutes)

A projectile launcher consists of a spring with an attached plate, as shown in Figure 1. When the spring is compressed, the plate can be held in place by a pin at any of three positions *A*, *B*, or *C*. For example, Figure 2 shows a steel sphere placed against the plate, which is held in place by a pin at position *C*. The sphere is launched upon release of the pin.

A student hypothesizes that the spring constant of the spring inside the launcher has the same value for different compression distances.

(a) The student plans to test the hypothesis by launching the sphere using the launcher.

i. State a basic physics principle or law the student could use in designing an experiment to test the hypothesis.



Students, write your response!

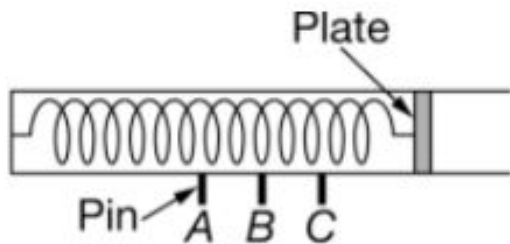


Figure 1. Uncompressed spring

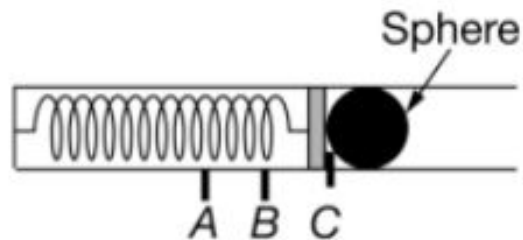


Figure 2. Compressed spring

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A student hypothesizes that the spring constant of the spring inside the launcher has the same value for different compression distances.

Conservation of Energy

(a) The student plans to test the hypothesis by launching the sphere using the launcher.

i. State a basic physics principle or law the student could use in designing an experiment to test the hypothesis.

A student hypothesizes that the spring constant of the spring inside the launcher has the same value for different compression distances.

ii. Using the principle or law stated in part (a)(i), determine an expression for the spring constant in terms of quantities that can be obtained from measurements made with equipment usually found in a school physics laboratory.

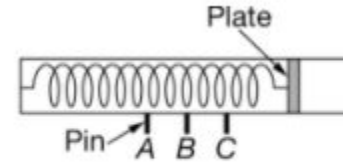


Figure 1. Uncompressed spring

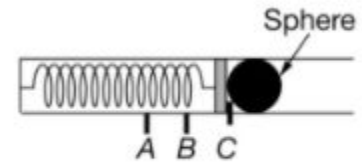


Figure 2. Compressed spring



Students, draw anywhere on this slide!

A student hypothesizes that the spring constant of the spring inside the launcher has the same value for different compression distances.

ii. Using the principle or law stated in part (a)(i), determine an expression for the spring constant in terms of quantities that can be obtained from measurements made with equipment usually found in a school physics laboratory.

$$U_s = K$$

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$k = \frac{mv^2}{x^2}$$

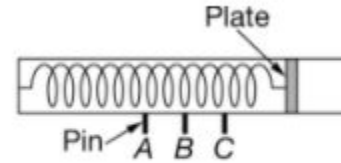


Figure 1. Uncompressed spring

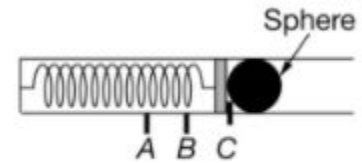


Figure 2. Compressed spring



The logo for fiveable, featuring the word "fiveable" in a bold, sans-serif font. "five" is white and "able" is dark blue. The text is contained within a light blue speech bubble shape that points downwards.

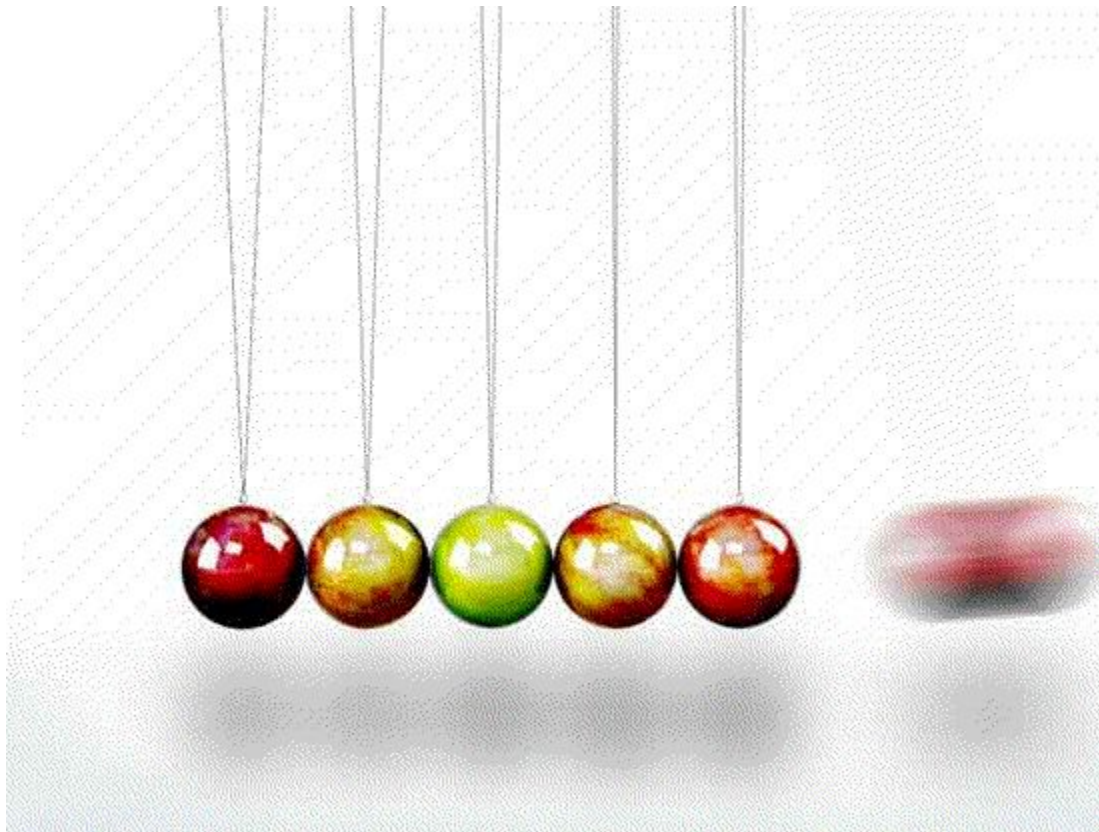
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AP Physics 1

Finale Part 2--Momentum!

- **5:00-5:55 Energy!**
- **6:00-6:55 Momentum!**
- **7:00-7:55 Simple Harmonic Motion**
- **8:00-8:55 FRQ Tips and Tricks**
- **9:00-9:00 MCQ Tips and Tricks
with GimKit Challenge!**

Momentum!



Translational (Linear)

$$\vec{p} = m\vec{v}$$

$$\Delta\vec{p} = \vec{F} \Delta t$$

Angular

$$L = I\omega$$

$$\Delta L = \tau \Delta t$$

BE CAREFUL! The following equation is for DENSITY--NOT momentum!!

$$\rho = \frac{m}{V}$$

density (greek letter rho) = mass / VOLUME (not velocity!)

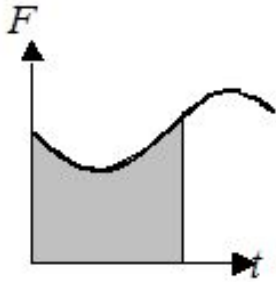
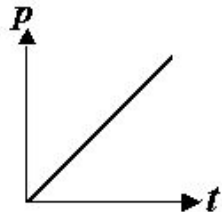
Important Quantities

Name	Symbol	Units	Basic Equation
Momentum	$\mathbf{p}$	kg·m/s	$\mathbf{p} = m\mathbf{v}$

Name	Symbol	Units	Basic Equation
Impulse	$\mathbf{J}$ or $\Delta\mathbf{p}$	kg·m/s	$\Delta\mathbf{p} = m\Delta\mathbf{v} = \mathbf{F}\Delta t$

Important Equations

Name	Equation	Given?	Notes
Conservation of Momentum	$\sum \mathbf{p}_i = \sum \mathbf{p}_f$	No	Use if you are given any problem dealing with a collision. Keep in mind that momentum is a vector, so if an object travels backwards, then it has negative momentum.

Name	Graph (Shape)	Notes
Force vs. Time	 (Could be anything)	The area under a force vs. time is equal to impulse. Impulse is the change in the momentum of the object the force is applied to. If there is a collision of two objects, the impulse is applied to both objects, but in opposite directions (Newton's Third Law).
Momentum vs. Time		The slope of momentum vs. time is net force, because $\mathbf{F}_{net} = \frac{\Delta \mathbf{p}}{\Delta t}$

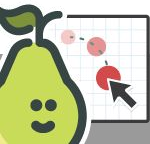
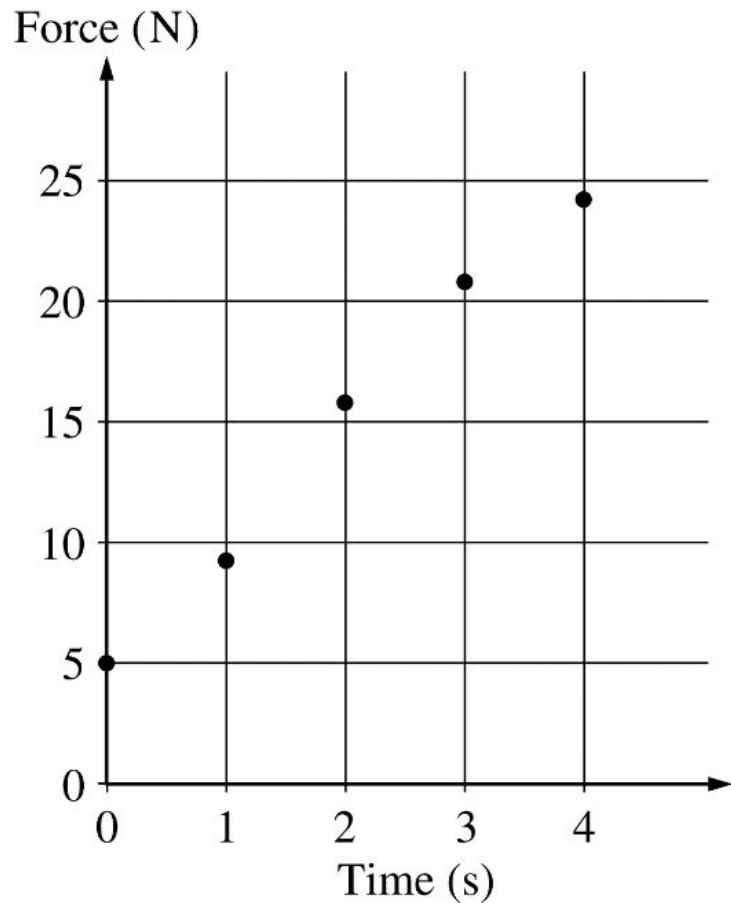
Important Graphs

A student obtains data on the magnitude of force applied to an object as a function of time and displays the data on the graph above.

Answer the following question for the information and diagram above.

The increase in the momentum of the object between $t = 0$ s and $t = 4$ s is most nearly

- A. 40 N·s
- B. 50 N·s
- C. 60 N·s
- D. 80 N·s



Students, drag the icon!



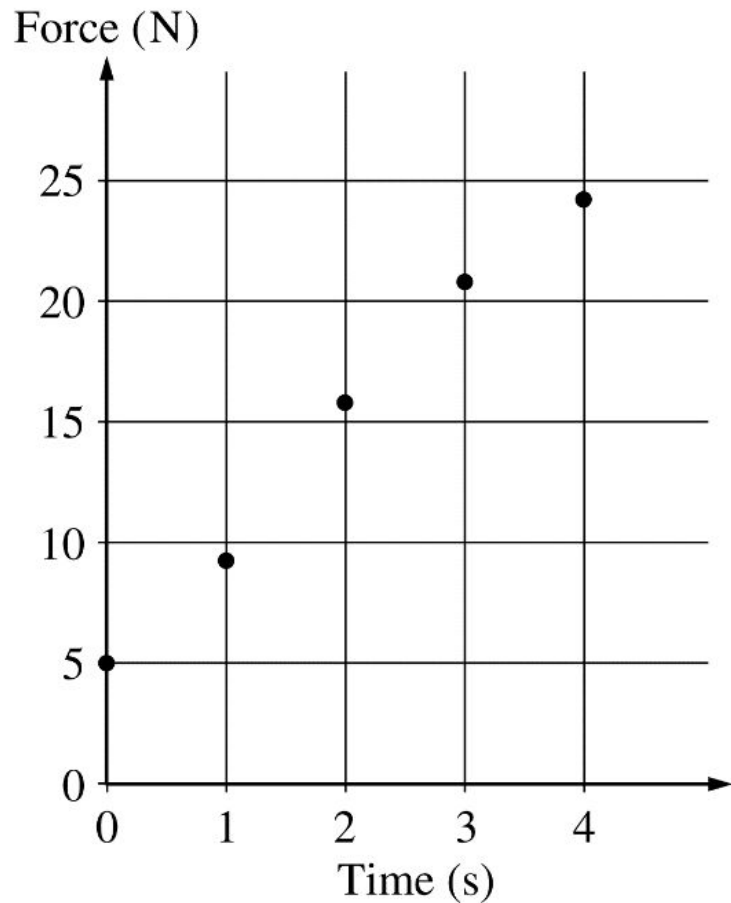
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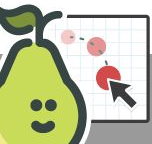
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A force of constant magnitude F and fixed direction acts on an object of mass m that is initially at rest. If the force acts for a time interval Δt over a displacement Δx , what is the magnitude of the resultant change in the linear momentum of the object?

- A. $F \Delta t$
- B. $F \Delta x$
- C. $F \Delta t / m$
- D. $F \Delta x / m$



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$$\text{Impulse} = F_{\text{net}} \Delta t$$

- Units = $\text{N} \cdot \text{s} = \text{kg} \cdot \text{m/s}^2 \cdot \text{s} = \text{kg} \cdot \text{m/s}$

$$\text{Momentum} = m \Delta v$$

- Units = $\text{kg} \cdot \text{m/s}$

Impulse-Momentum Theorem

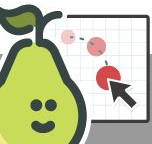
$$F_{\text{net}} \Delta t = (m \Delta v)$$

For Airbags, Padded Dashboards, Helmets, etc...

$$F \Delta t = F_{\Delta t}$$

How does an air mattress protect a stunt person landing on the ground after a stunt?

- A. It reduces the momentum change of the stunt person.
- B. It increases the momentum change of the stunt person.
- C. It shortens the stopping time of the stunt person and increases the force applied during the landing.
- D. It lengthens the stopping time of the stunt person and reduces the force applied during the landing.



How does an air mattress protect a stunt person landing on the ground after a stunt?

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- C. It shortens the stopping time of the stunt person and increases the force applied during the landing.
- D. It lengthens the stopping time of the stunt person and reduces the force applied during the landing.**

Big concept: Use momentum to solve collision problems!

- USE CONSERVATION OF MOMENTUM WHENEVER THERE IS ANY KIND OF COLLISION! DO NOT USE CONSERVATION OF ENERGY!
- DON'T ASSUME that you know that a collision is elastic or inelastic unless you are told.
- DON'T ASSUME that the objects stick together unless you are told.
- When setting up a conservation of momentum equation, remember that a velocity is negative if the object is moving to the left (or down).

Important Concepts

Stick, bounce, explosion?	Collision Type	Conservation of Momentum Equation		Kinetic Energy Relationship
		Initial	Final	
Explosion	Explosion	$(m_1 + m_2)v_o = m_1v_1 + m_2v_2$		$K_i < K_f$
Bounce	Elastic	$m_1v_1 + m_2v_2 = m_1v_1 + m_2v_2$		$K_i = K_f$
Bounce	Inelastic	$m_1v_1 + m_2v_2 = m_1v_1 + m_2v_2$		$K_i > K_f$
Stick	Perfectly Inelastic	$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$		$K_i > K_f$

To review:

- An elastic collision is one in which no kinetic energy is lost to heat or deformation. In these problems, you must also set up a conservation of kinetic energy equation.
- An inelastic collision is one in which some kinetic energy is lost during the collision, but the objects may not stick together. In other words, total kinetic energy before the collision is more than total kinetic energy after the collision.
- A perfectly inelastic collision is one in which the two objects stick together. The most possible energy is lost to heat and deformation during a perfectly inelastic collision.

Elastic Collisions

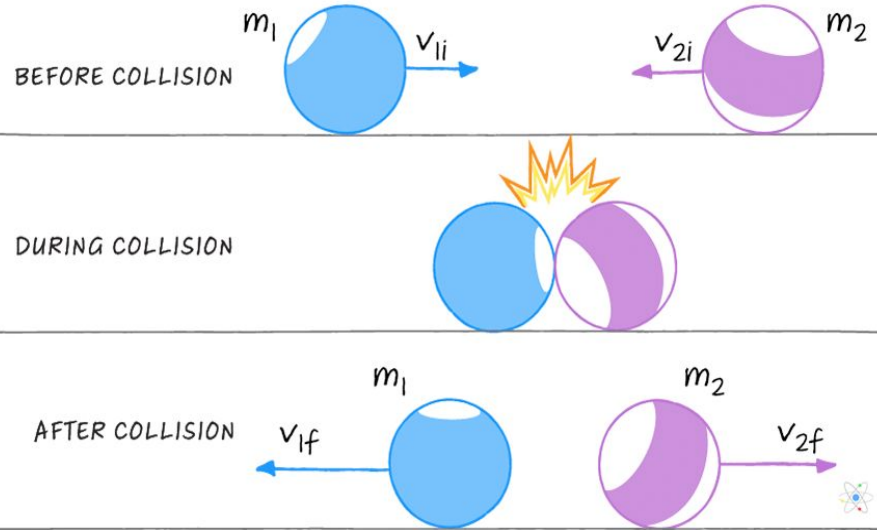
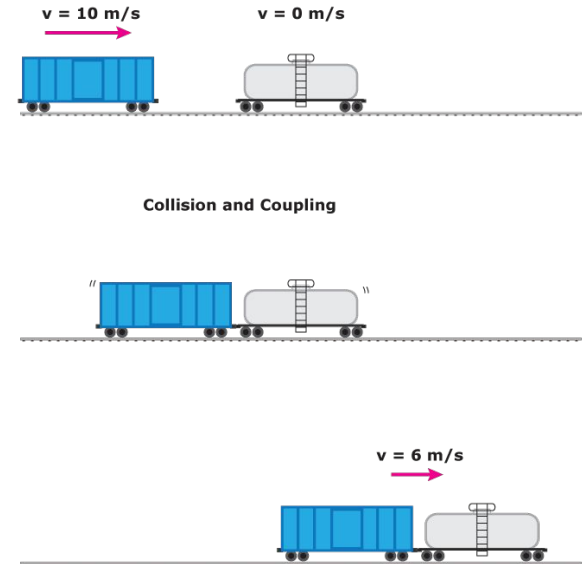


Image from [Sciencing](#)

"Perfectly Inelastic" Collisions

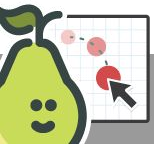


Important Concepts



A railroad car of mass m is moving with speed u when it collides with and connects to a second railroad car of mass $3m$, initially at rest, as shown above. How do the speed and kinetic energy of the connected cars compare to those of the single car of mass m before the collision?

Answer Choice	Speed	Kinetic Energy
A.	Less	Less
B.	Less	The same
C.	The same	Less
D.	The same	The same



Important Concepts



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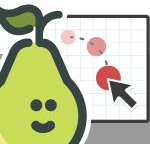
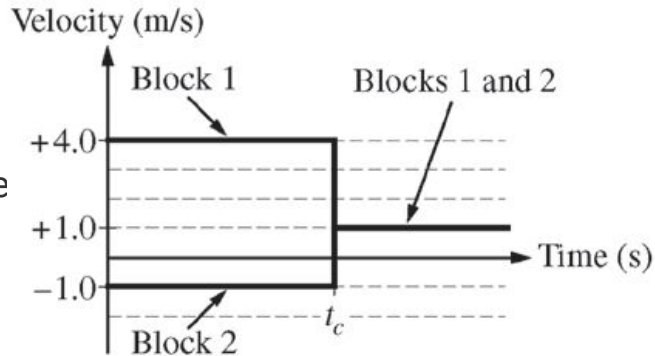
Answer Choice	Speed	Kinetic Energy
A.	Less	Less
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C.	The same	Less
D.	The same	The same

Important Concepts

Block 1 of mass m_1 and block 2 of mass m_2 are sliding along the same line on a horizontal frictionless surface when they collide at time t_c . The graph above shows the velocities of the blocks as a function of time.

How does the kinetic energy of the two-block system after the collision compare with its kinetic energy before the collision, and why?

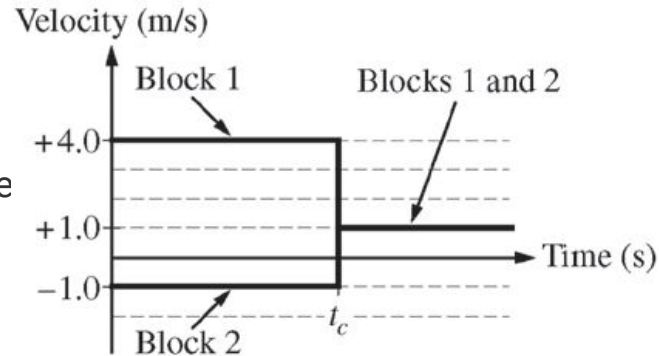
- A. It is less, because the blocks have the same velocity after the collision, so some of their kinetic energy was transformed into internal energy.
- B. It is less, because the blocks have the same velocity after the collision, so some of their kinetic energy was transformed into internal energy.
- C. It is less, because the blocks have velocities in opposite directions before the collision, so some of their kinetic energy cancels.
- D. It is the same, because the collision was instantaneous, so the effect of external forces during the collision is negligible.



Important Concepts

Block 1 of mass m_1 and block 2 of mass m_2 are sliding along the same line on a horizontal frictionless surface when they collide at time t_c . The graph above shows the velocities of the blocks as a function of time.

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- D. It is the same, because the collision was instantaneous, so the effect of external forces during the collision is negligible.

During a collision (or other interaction), momentum is often conserved (meaning that the net momentum of the two objects before the collision equals the net momentum after the collision). Here is what must be true in order for momentum to be conserved:

- EITHER all the external forces (forces on objects in the system from objects outside the system) acting on the system balance,
- OR, the “before” instant and the “after” instant are very close in time, so that external forces don’t have enough time to change the system’s momentum (using $\Delta p = F_{\text{net}} \cdot \Delta t$)

The most common way to write an equation for Conservation of Momentum is:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

If two objects stick together as a result of the collision, we treat them as a single object after the collision:

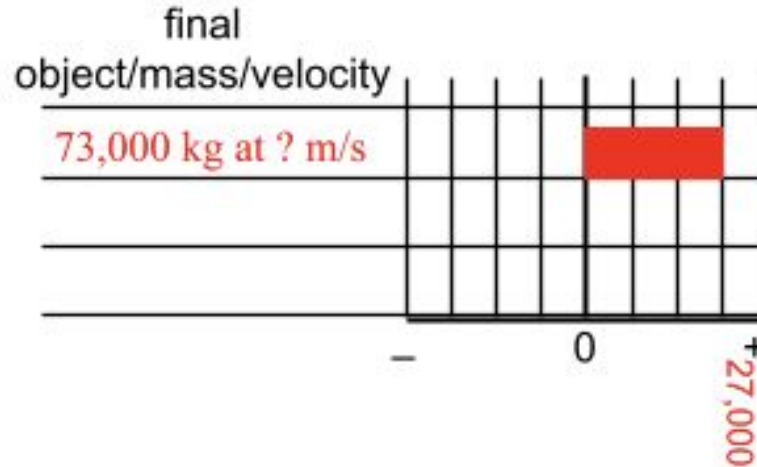
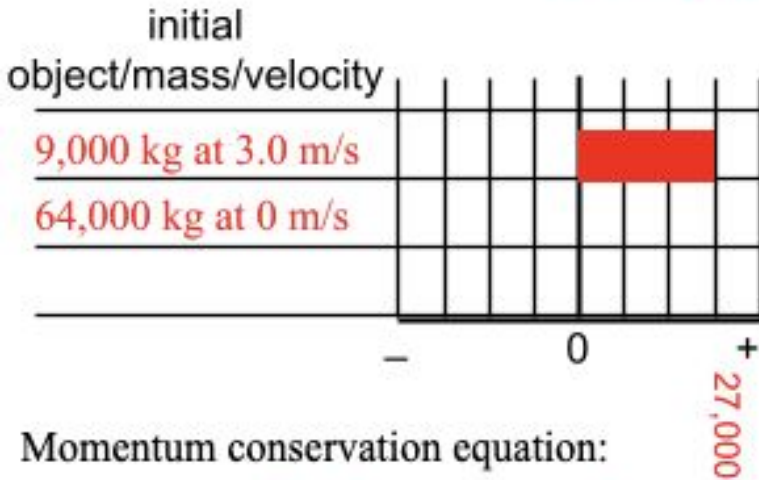
$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

Likewise, if two objects start stuck together and “explode” apart, we treat them as a single object before the collision:

$$(m_1 + m_2) v_i = m_1 v_{1f} + m_2 v_{2f}$$

In a railroad yard, a train is being assembled. An empty boxcar, coasting at 3.0 m/s, strikes a loaded car that is stationary, and the cars couple together. Each of the boxcars has a mass of 9000 kg when empty, and the loaded car contains 55,000 kg of lumber.

event: **A totally inelastic collision.**



Momentum conservation equation:

$$m_{9000}v_{9000i} + 0 = (m_{9000} + m_{64,000})v_f$$

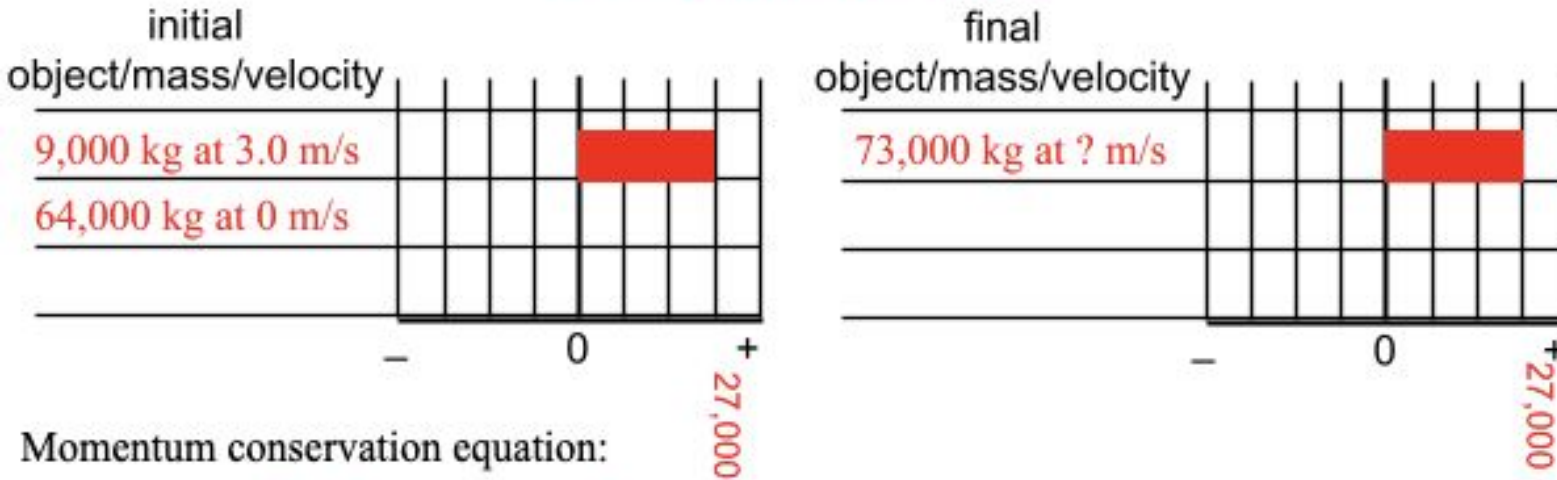
Find the speed of the coupled cars.



Students, draw anywhere on this slide!

In a railroad yard, a train is being assembled. An empty boxcar, coasting at 3.0 m/s, strikes a loaded car that is stationary, and the cars couple together. Each of the boxcars has a mass of 9000 kg when empty, and the loaded car contains 55,000 kg of lumber.

event: A totally inelastic collision.

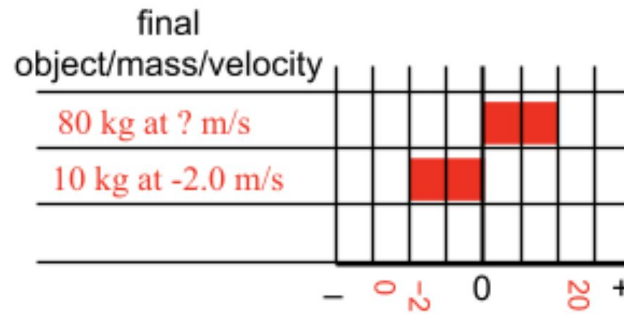
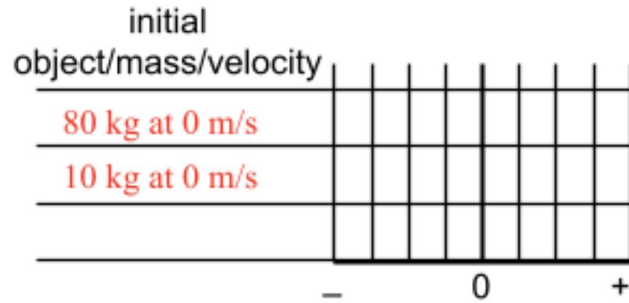


Momentum conservation equation:

$$v_f = \frac{m_{9000i} v_{9000i}}{(m_{9000} + m_{64,000})} \Rightarrow v_f = \frac{9,000\text{kg}(3.0 \frac{\text{m}}{\text{s}})}{(9,000\text{kg} + 64,000\text{kg})} = 0.37 \frac{\text{m}}{\text{s}}$$

An astronaut of mass 80 kg carries an empty oxygen tank of mass 10 kg. By pushing the tank away with a speed of 2.0 m/s, the astronaut recoils in the opposite direction. Complete the momentum conservation diagram.

event: **EXPLOSION**



$$m_{80i}v_{80i} + m_{10i}v_{10i} = m_{80f}v_{80f} + m_{10f}v_{10f}$$

b. Momentum conservation equation:

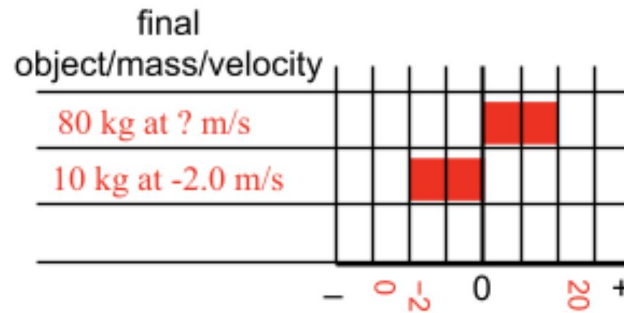
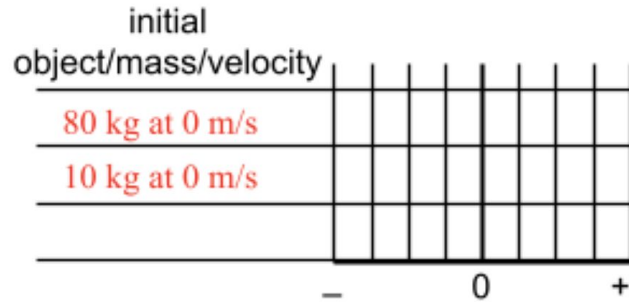
Find the speed with which the astronaut moves off into space.



Students, draw anywhere on this slide!

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event: **EXPLOSION**



b. Momentum conservation equation:

$$m_{80i}v_{80i} + m_{10i}v_{10i} = m_{80f}v_{80f} + m_{10f}v_{10f}$$

Find the speed with which the astronaut moves off into space.

$$0 + 0 = 80\text{kg}(v_{80f}) + 10\text{kg}(-2.0\frac{\text{m}}{\text{s}}) \Rightarrow v_{80f} = \frac{20\frac{\text{kg}\cdot\text{m}}{\text{s}}}{80\text{kg}} = 0.25\frac{\text{m}}{\text{s}}$$



The logo for fiveable, featuring the word "fiveable" in a bold, sans-serif font. The word is split into "five" in white and "able" in dark blue, set against a light blue speech bubble background.

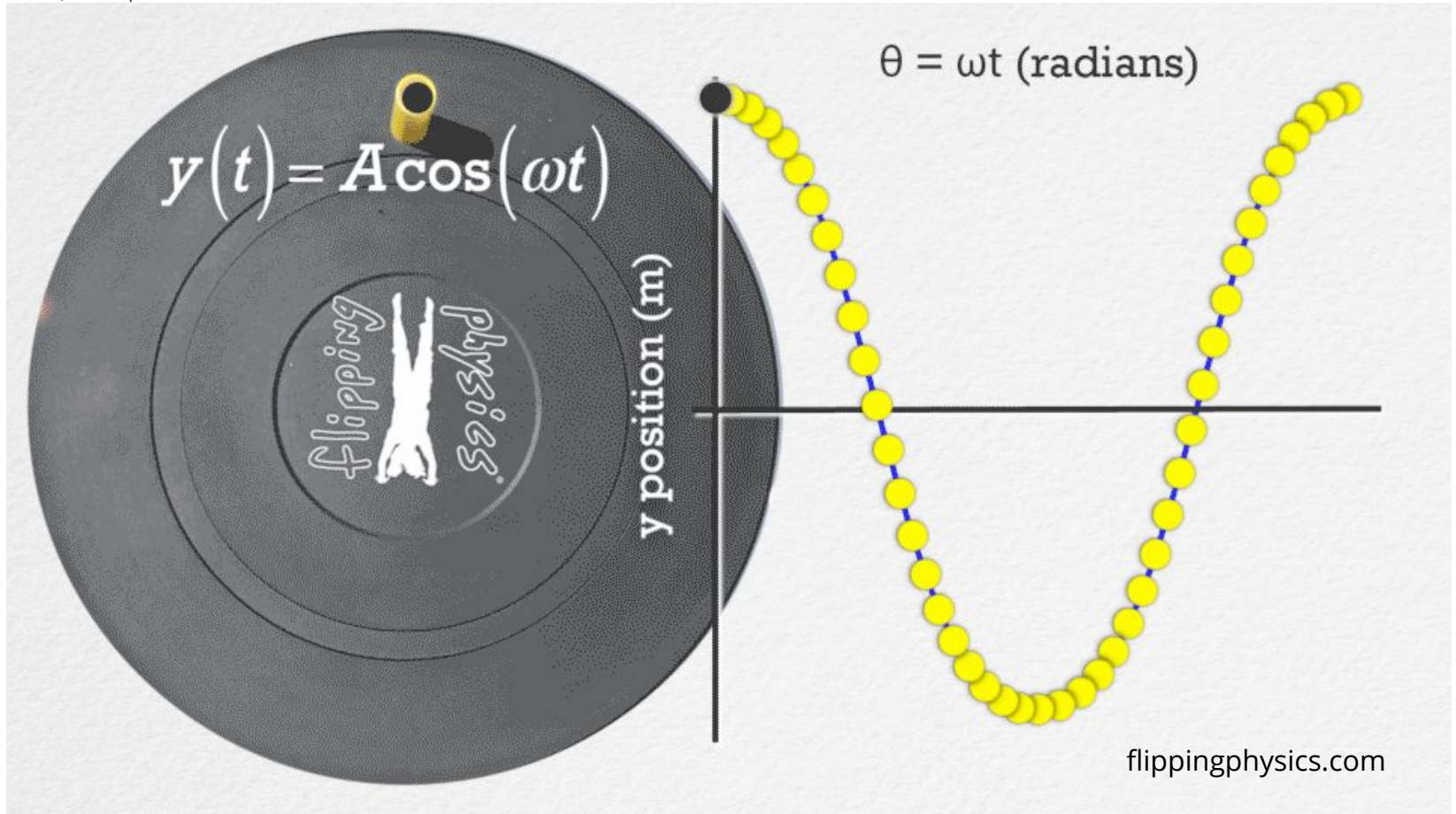
fiveable

AP Physics 1

Finale Part 3--Simple Harmonic Motion!

- **5:00-5:55 Energy!**
- **6:00-6:55 Momentum!**
- **7:00-7:55 Simple Harmonic Motion**
- **8:00-8:55 FRQ Tips and Tricks**
- **9:00-9:00 MCQ Tips and Tricks
with GimKit Challenge!**

Simple Harmonic Motion



$$x = A \cos(2\pi ft)$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

Important Quantities

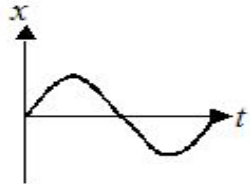
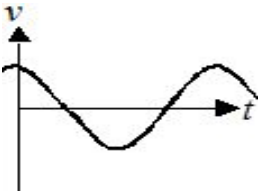
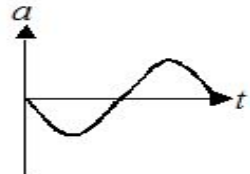
Name	Symbol	Units	Basic Equation
Amplitude	A	m	None

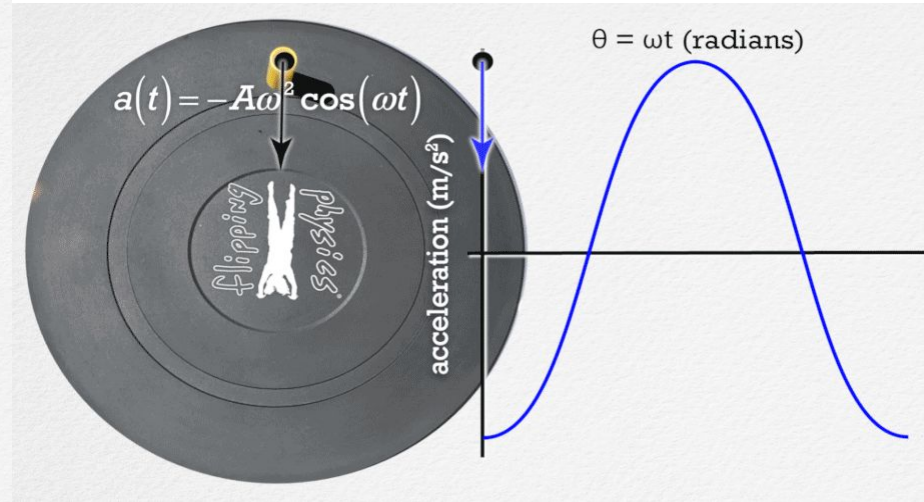
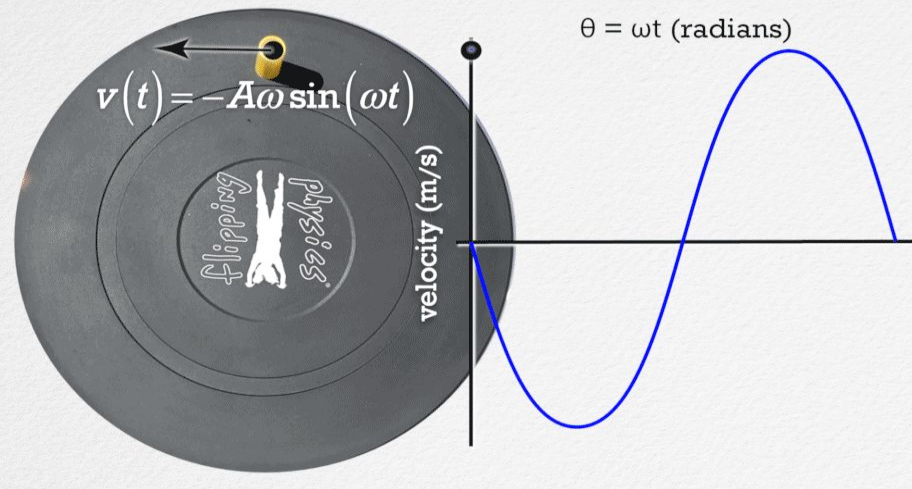
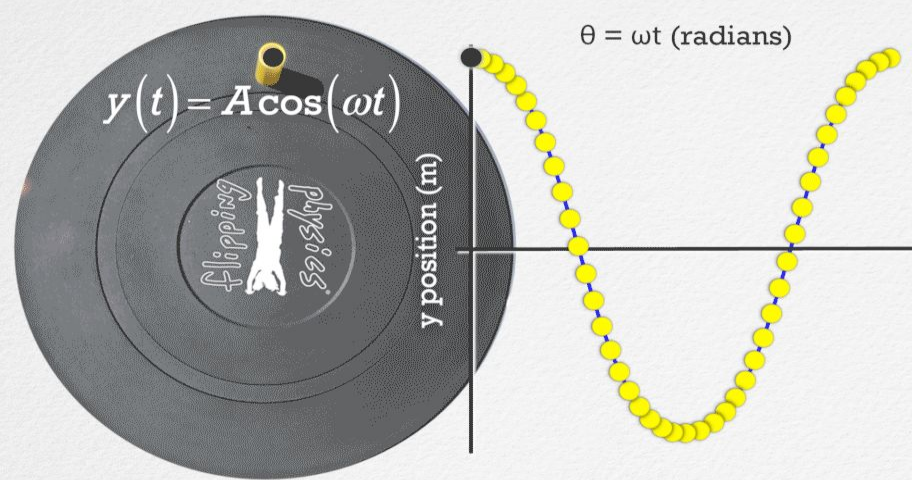
Name	Symbol	Units	Basic Equation
Spring Constant	k	N/m	$F_s = kx$

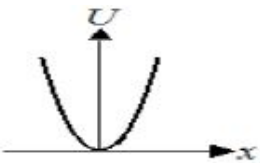
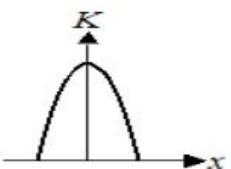
Important Equations

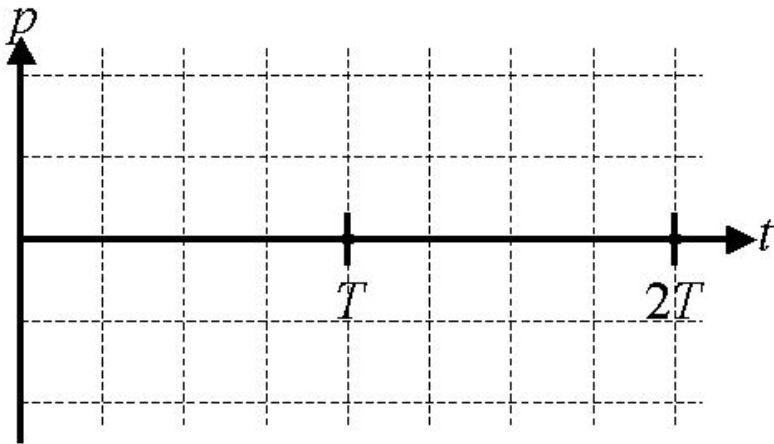
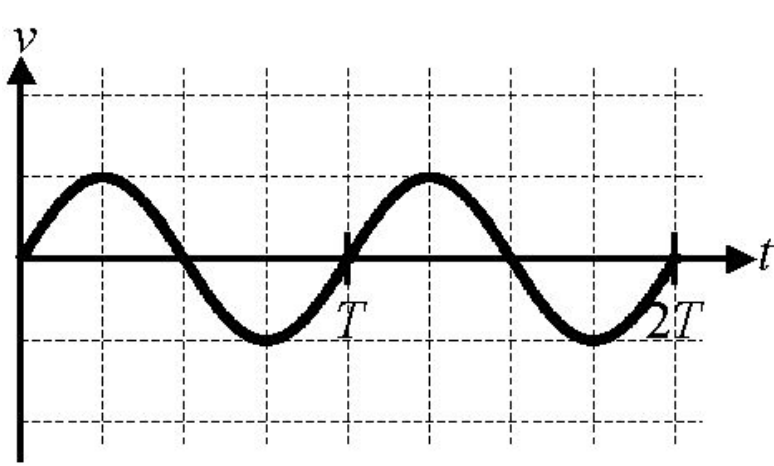
Name	Equation	Given ?	Notes
Period of a Spring-Mass Oscillator	$T_s = 2\pi\sqrt{\frac{m}{k}}$	Yes	A larger mass will make the period longer, but a stiffer spring will make the period shorter.
Period of a Pendulum	$T_p = 2\pi\sqrt{\frac{\ell}{g}}$	Yes	The mass at the end of the pendulum does not affect period. A longer string results in a longer period.
Angular Frequency of a Spring-Mass Oscillator	$\omega = \sqrt{\frac{k}{m}}$	No	Don't memorize this. Instead remember the above equation for period and know that. $\omega = \frac{2\pi}{T}$
Total Energy for a Spring-Block Oscillator	$E_{tot} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$	No	Total energy is constant throughout the entire oscillation, but "trades off" between kinetic and potential.

Important Graphs

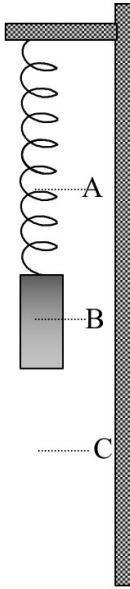
Name	Graph (Shape)	Notes
Position vs. Time		Displacement varies sinusoidally with time. The highest peak and lowest trough is $x = \pm A$ Maximum displacement occurs at the endpoints of oscillation.
Velocity vs. Time		Velocity varies sinusoidally with time. The highest peak and lowest trough is $v = \pm A\omega$ Maximum velocity occurs at the equilibrium point.
Acceleration vs. Time		Acceleration varies sinusoidally with time. The highest peak and lowest trough is $a = \pm A\omega^2$ Maximum acceleration occurs at the endpoints.



Potential Energy vs. Time		The maximum potential energy is $U_{\max} = \frac{1}{2}kA^2$ and only occurs at the points $x = A$ and $x = -A$
Kinetic Energy vs. Time		The maximum kinetic energy is $K_{\max} = \frac{1}{2}kA^2$, and only occurs at the point.
Potential Energy vs. Displacement		PAY ATTENTION TO THE AXES! THIS HAS DISPLACEMENT ON THE HORIZONTAL!
Kinetic Energy vs. Displacement		PAY ATTENTION TO THE AXES! THIS HAS DISPLACEMENT ON THE HORIZONTAL!



A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of momentum vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.

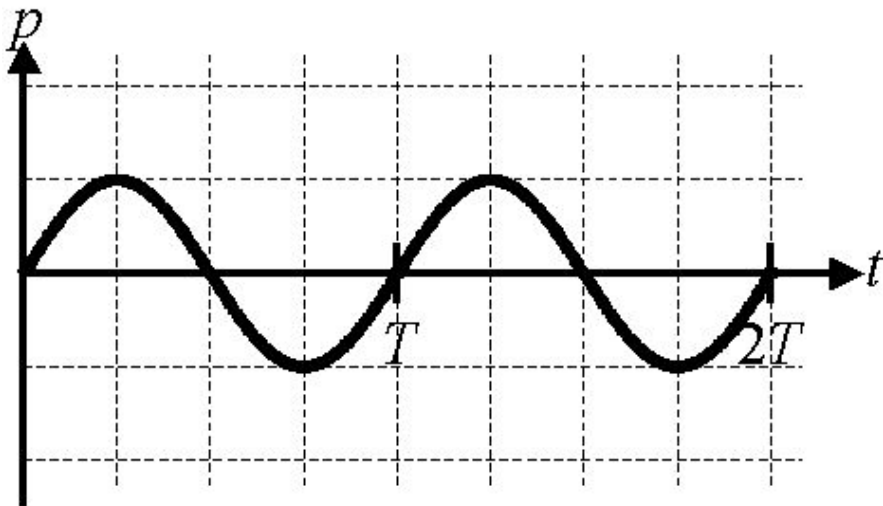
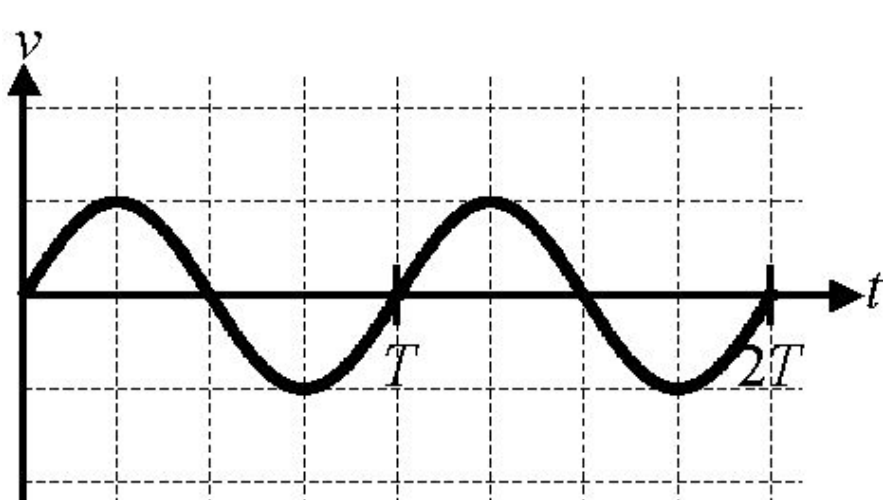


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and AMTA Modeling SHM

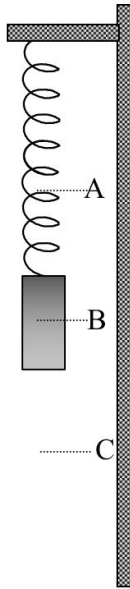


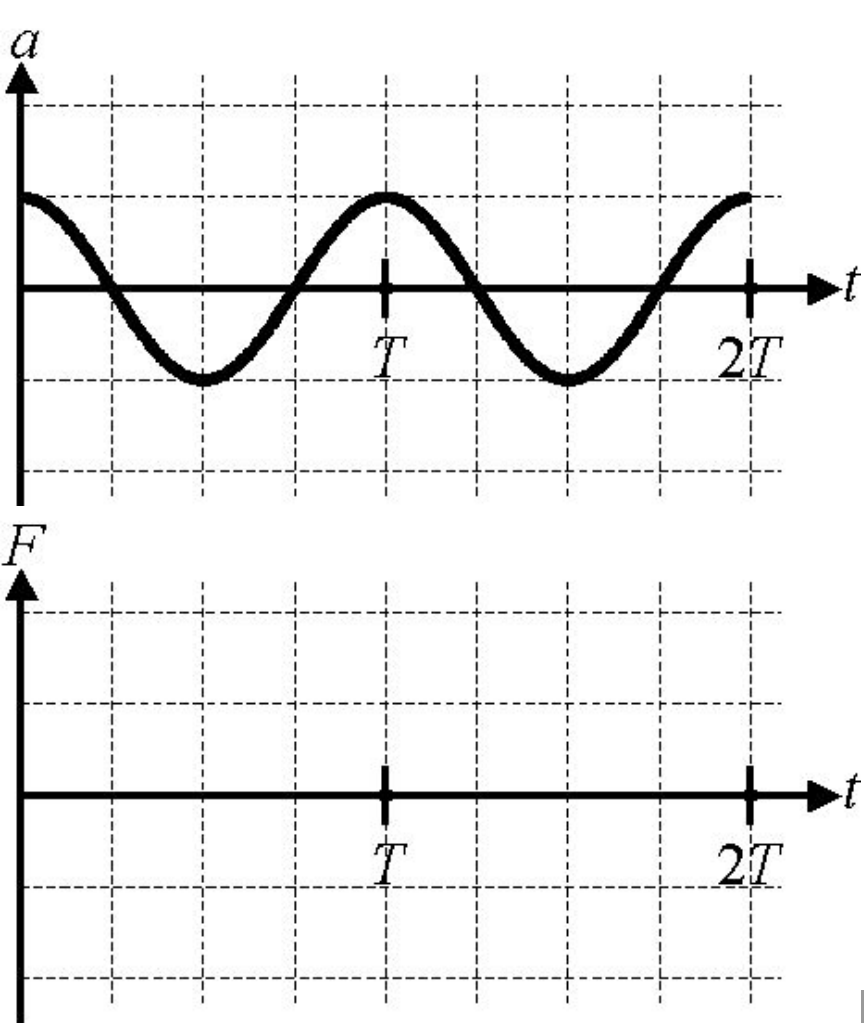
Students, draw anywhere on this slide!

Pear Deck Interactive Slide
Do not remove this bar

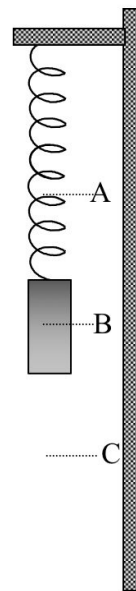


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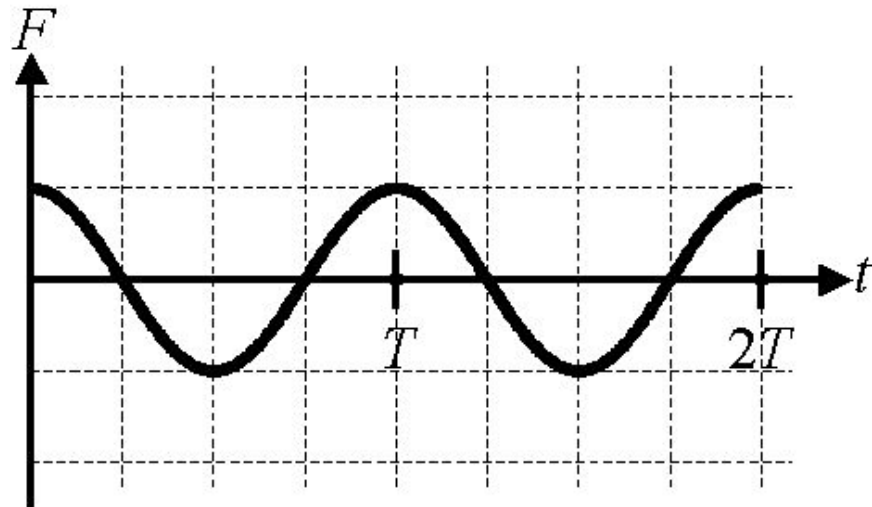
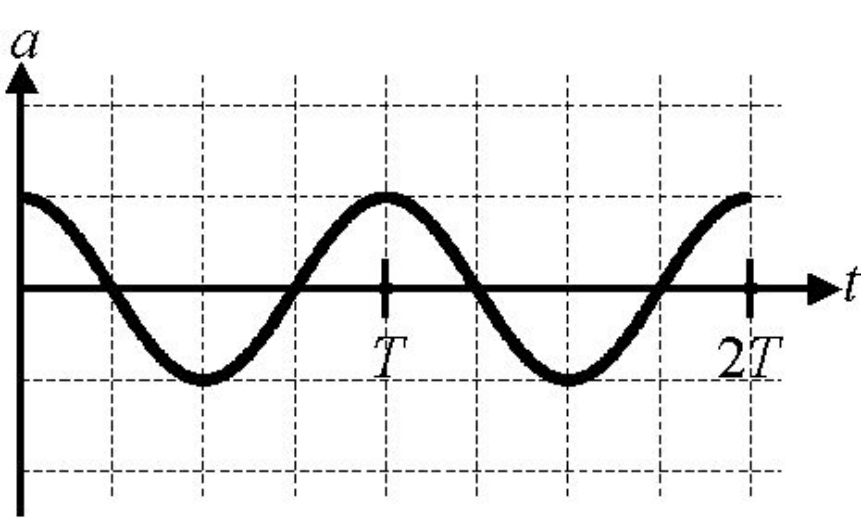




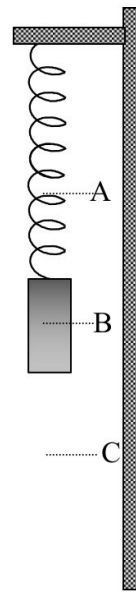
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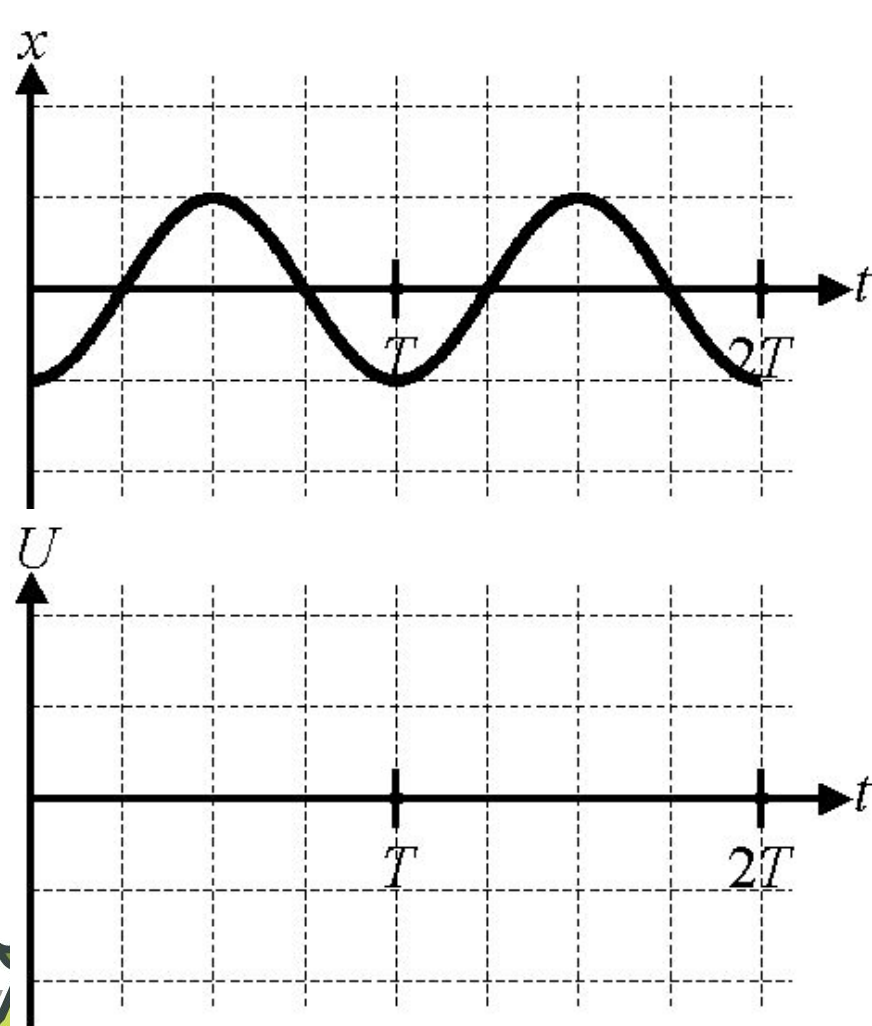


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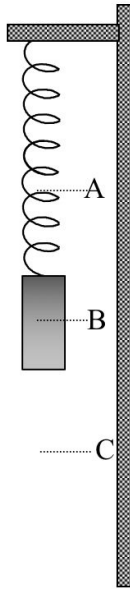


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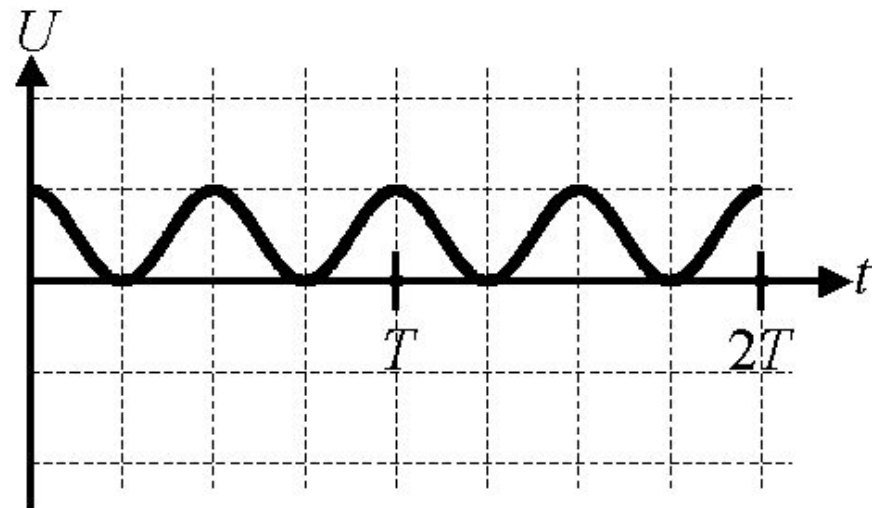
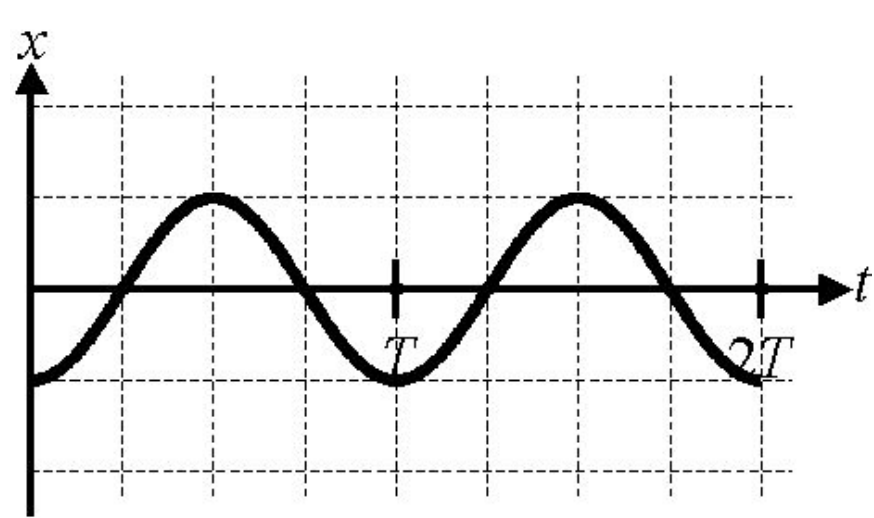




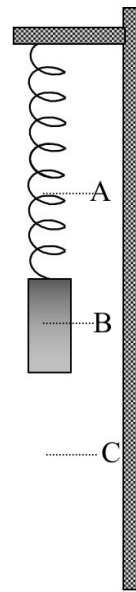
A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of potential energy vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.

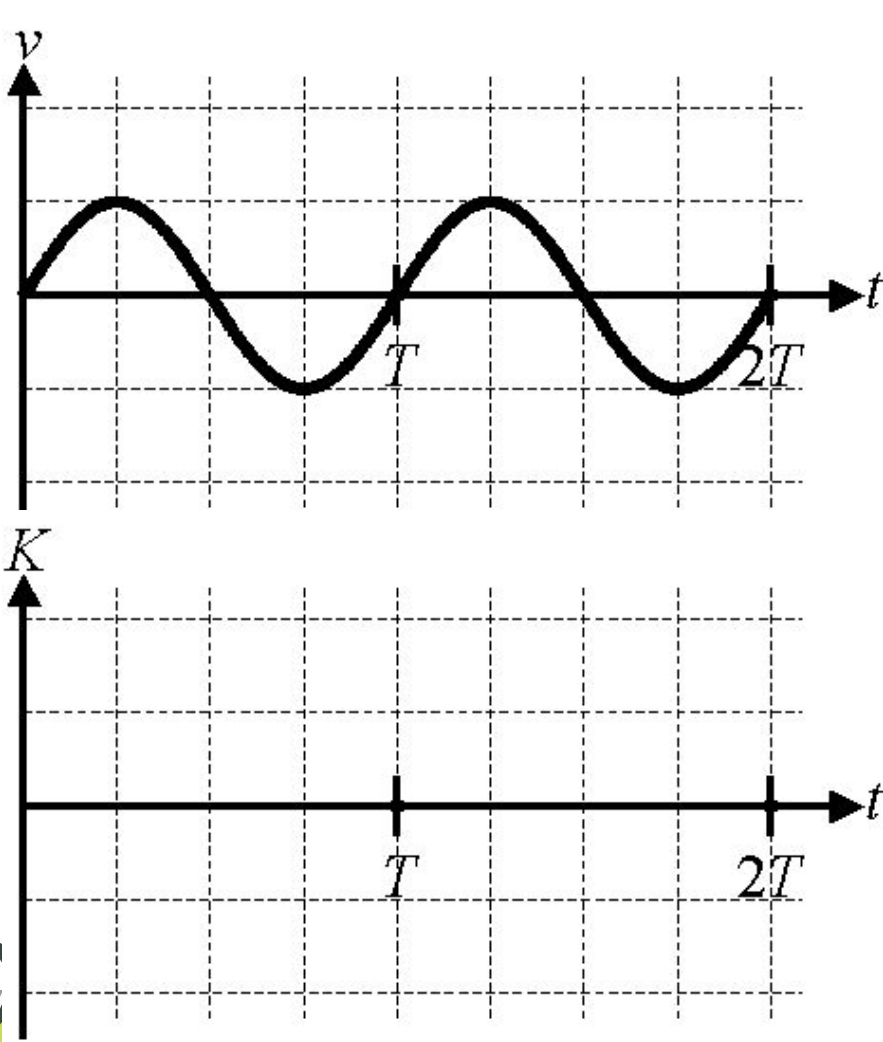


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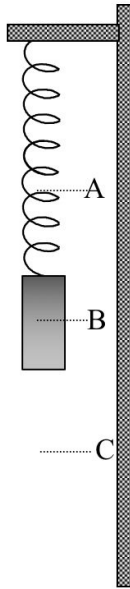


A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of potential energy vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.

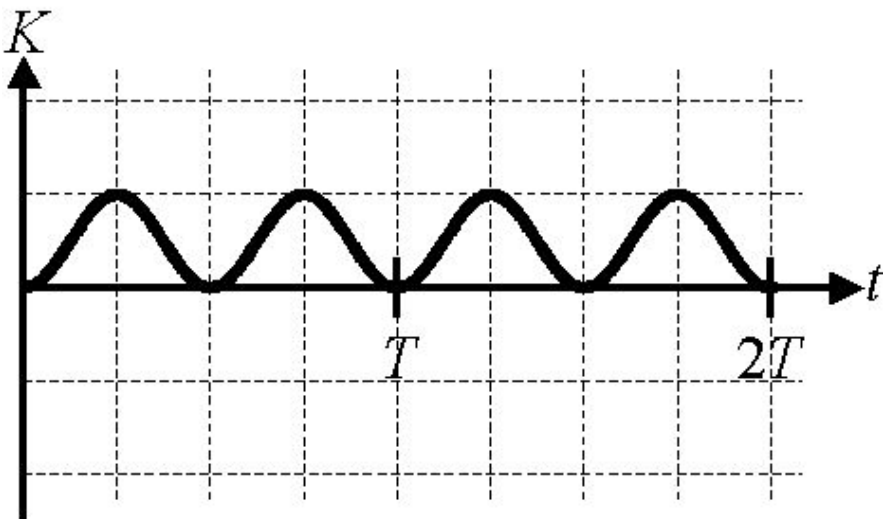
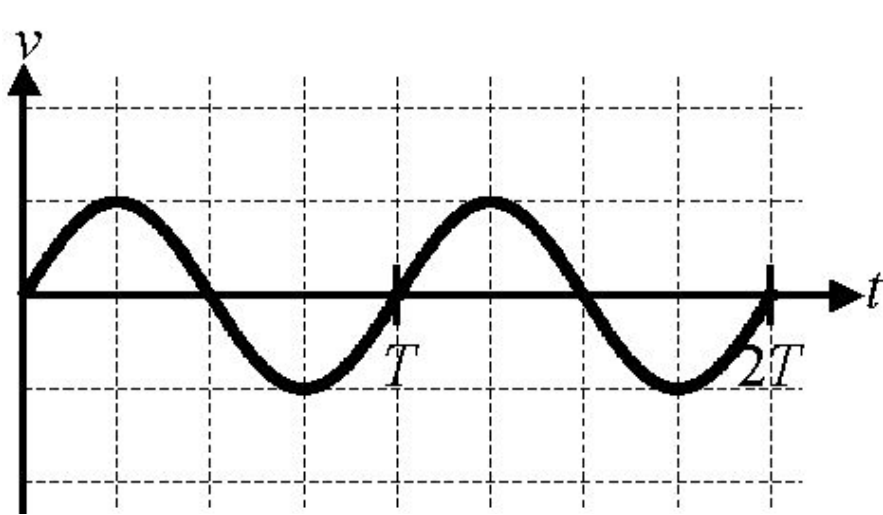




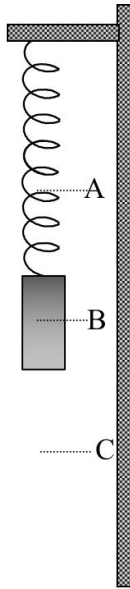
A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of kinetic energy vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.



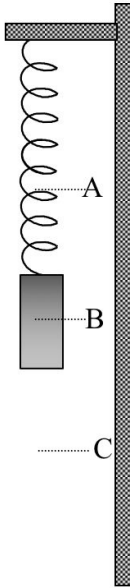
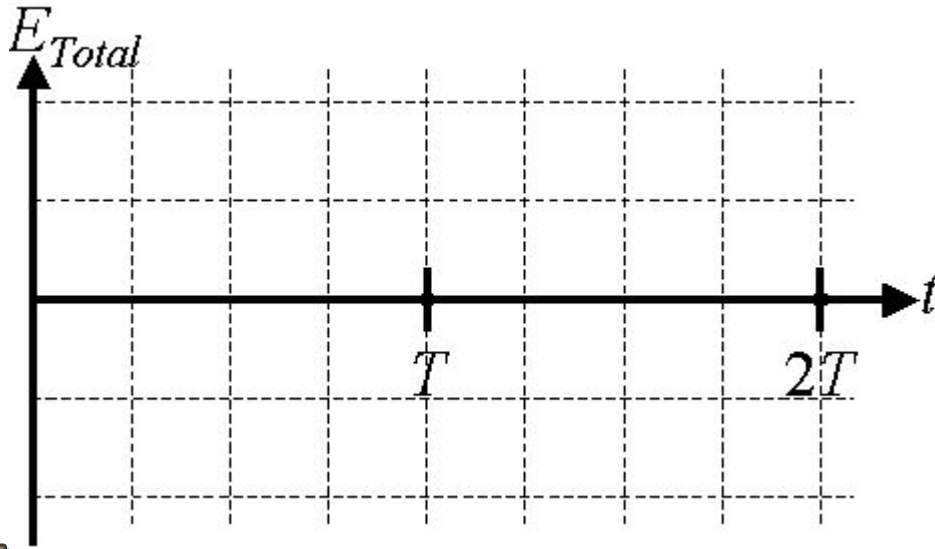
Images from NMSI SHM Review SSS Packet and AMTA Modeling SHM



A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of kinetic energy vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.



A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of total energy vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.

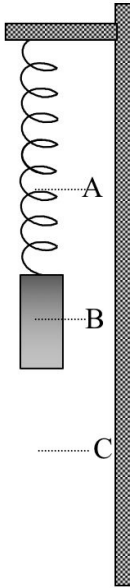
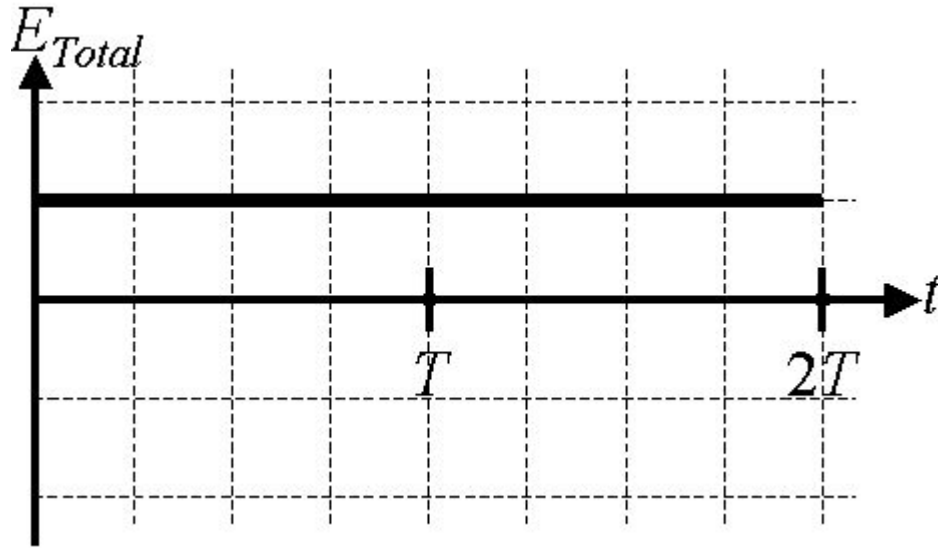


Images from NMSI SHM Review SSS Packet
and AMTA Modeling SHM

Students, draw anywhere on this slide!

Pear Deck Interactive Slide
Do not remove this bar

A block hangs from a spring. The block is pulled down a small distance from its equilibrium position and released at time $t = 0$. The period of the resulting simple harmonic motion is T . Sketch a graph of total energy vs. period (T) for this block-spring system. Let vector quantities be positive for the upward direction.



Images from NMSI SHM Review SSS Packet
and AMTA Modeling SHM

Periodic Motion

Describes any motion that repeats over and over the same way.

- Uniform circular motions
- Simple harmonic motions
- Orbits (circular or elliptical)
 - If a puck on an air hockey table could bounce back and forth between both sides forever without losing energy, that would be periodic motion.
 - If a golf ball would bounce on the floor back up to the same height forever without losing energy, that would be periodic motion.

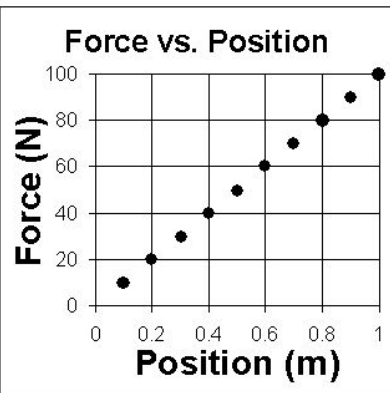
Important Concepts

- For an object to exhibit periodic motion in one dimension:
 - There must be a location called “equilibrium” where the object experiences no net force.
 - There must be a net force acting on the object that is always directed toward equilibrium.

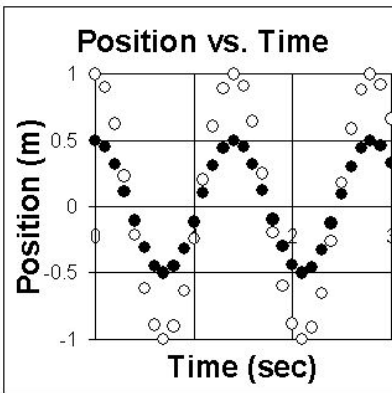
Important Concepts

- Simple harmonic motion is “simple” because it is the periodic motion that is the simplest to model using mathematics.
- If an object exhibits simple harmonic motion, then the following three properties are true for that object. If one of the properties is true, they are all true together (if one is false, they are all false together), and the object exhibits SHM:
 - The net force on the object is directly proportional to the object's displacement from equilibrium.
 - The object's motion can be modeled as a single sine or cosine graph (or function).
 - The period of the motion is independent of the amplitude of the oscillation.

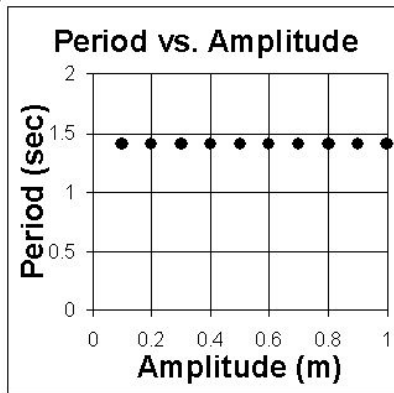
A spring-and-mass meets all 3 criteria...



A spring exerts force according to the equation $F_s = kx$. That means that a spring exerts force that is directly proportional to displacement.



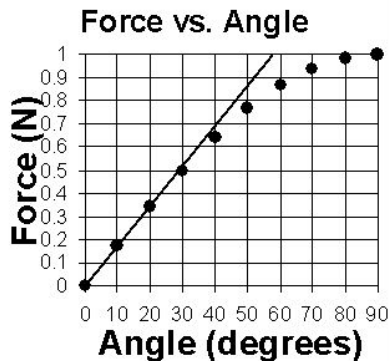
The graph of the position vs. time of an object oscillating on the end of a spring is a sine or cosine function. Note that the period is the same regardless of amplitude.



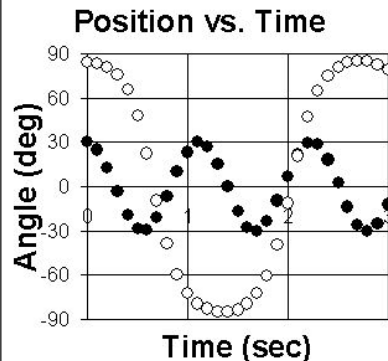
The period does not depend on the amplitude of the motion. Whether the motions are small or large, the period of the motions are the same.

A pendulum meets all three criteria ...

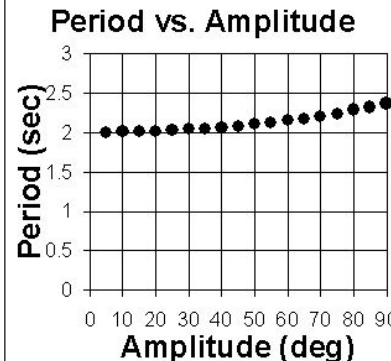
***if the pendulum does not swing more than $\sim 30^\circ$ from the vertical.**



For a pendulum, the net force on the bob follows a line as a function of displacement angle up to $\sim 30^\circ$. Beyond 30° , force is not proportional to displacement.

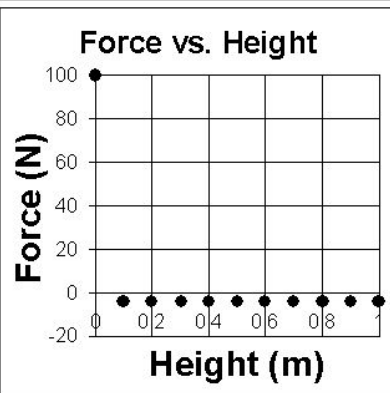


For amplitudes $\sim 30^\circ$ or less, angle vs. time is a sine or cosine. Above this amplitude, the function “flattens” at the peaks, which is not sine or cosine and not SHM.

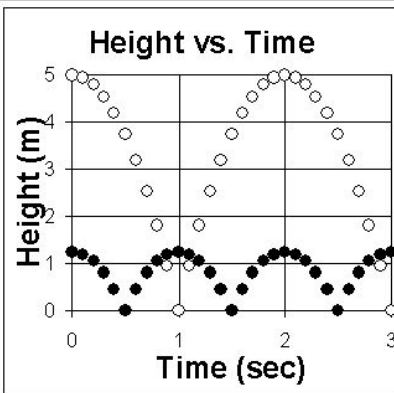


For angles less than 30° , the period is approximately constant with amplitude. Greater than this angle, period increases with amplitude, which is not SHM.

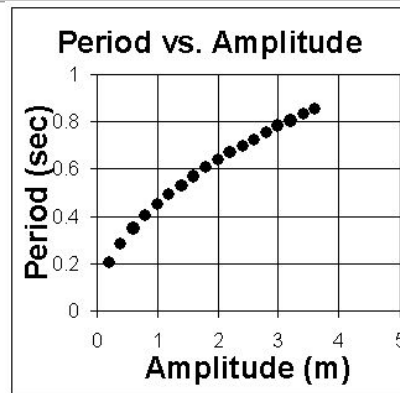
A bouncing golf ball does NOT meet any of the criteria--it is NOT simple harmonic motion



The force on the ball is huge when it hits the ground, then a constant $-mg$ when the ball is in the air. This is not a linear relationship.



The graph of height vs. time for a bouncing ball is NOT a sine or cosine graph. This means that a bouncing golf ball is not SHM.

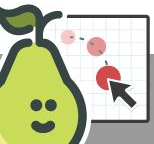


Period definitely depends on amplitude. Drop a ball from a low height, it bounces often, but from a high height, it bounces less often.

Important Concepts

A simple pendulum consisting of a small object of mass m attached to a string of length l has a period T . A pendulum with which of the following combinations of object mass and string length will also have period T ?

Answer Choice	Object Mass	String Length
A.	$m/2$	l
B.	m	$l/4$
C.	$\sqrt{2} m$	$l/\sqrt{2}$
D.	$2m$	$4l$



Important Concepts

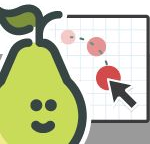
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D.	$2m$	$4l$

Important Concepts

A simple pendulum and a mass hanging on a spring both have a period of 1 s when set into small oscillatory motion on Earth. They are taken to Planet X, which has the same diameter as Earth but twice the mass. Which of the following statements is true about the periods of the two objects on Planet X compared to their periods on Earth?

- A. Both are the same.
- B. Both are longer.
- C. The period of the mass on the spring is shorter; that of the pendulum is the same.
- D. The period of the mass on the spring is shorter; that of the pendulum is the same.



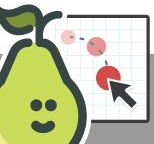
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- A. Both are the same.
- B. Both are longer.
- C. The period of the mass on the spring is shorter; that of the pendulum is the same.
- D. The period of the mass on the spring is shorter; that of the pendulum is the same.**

Important Concepts

A block attached to the lower end of a vertical spring oscillates up and down. If the spring obeys Hooke's law, the period of oscillation depends on which of the following?

- I. Mass of the block
 - II. Amplitude of the oscillation
 - III. Force constant of the spring
-
- A. II only
 - B. III only
 - C. I and II
 - D. I and III



A block attached to the lower end of a vertical spring oscillates up and down. If the spring obeys Hooke's law, the period of oscillation depends on which of the following?

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- A. II only
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- C. I and II
- D. I and III**



The logo for fiveable, featuring the word "fiveable" in a bold, sans-serif font. "five" is white and "able" is dark blue. The text is contained within a light blue speech bubble shape with a tail pointing downwards.

fiveable

AP Physics 1

Finale Part 4--FRQ Tips and Tricks!

- **5:00-5:55 Energy!**
- **6:00-6:55 Momentum!**
- **7:00-7:55 Simple Harmonic Motion**
- **8:00-8:55 FRQ Tips and Tricks**
- **9:00-9:00 MCQ Tips and Tricks
with GimKit Challenge!**

Free Response Question (FRQ) Tips and Tricks!

AP PHYSICS EXAM SECTION II

5 Free-Response Questions | 90 minutes

- Experimental Design 12 points $\approx$ 25 min
- Quantitative/Qualitative Translation 12 points $\approx$ 25 min
- Paragraph Length Response 7 points $\approx$ 12 min
- 2 Short Answer Questions 7 points $\approx$ 12 min each

Take a closer look at each FRQ to determine where to start. **Answer the questions that you feel confident about first!**

FRQ Tips and Tricks!--Key Terms!

"Calculate" means that you are expected to show **ALL** your work leading to your final answer: algebraic or numerical

"What is" and **"determine"** indicate that you do **not** need to show your work to obtain full credit. But, showing work leading to answers is a good idea because **partial credit** can be earned in the case of an incorrect answer

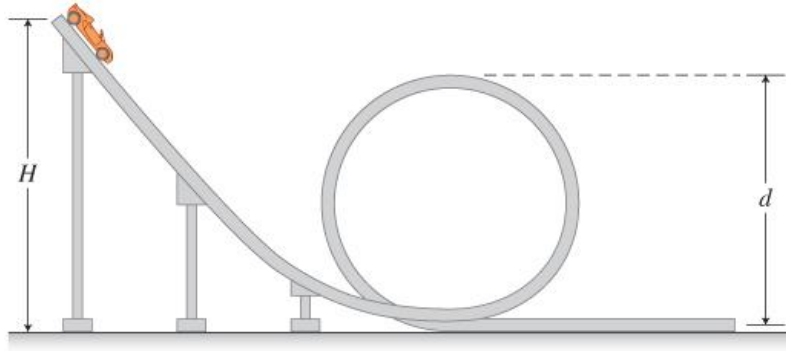
FRQ Tips and Tricks!--Key Terms!

"**Justify**" and "**explain**" call for an answer supported by prose, equations, calculations, diagrams, or graphs.

- The prose or equations may refer to **fundamental ideas** or relations in physics, such as Newton's laws, conservation of energy, conservation of momentum, etc.
- In other cases, the justification or explanation may take the form of **analyzing the behavior** of an equation for large or small values of a variable in the equation.

FRQ Tips and Tricks!--Key Terms!

"Derive" indicates that you need to begin your solution with one or more fundamental equations or physics principles. Show all the steps leading to your final answer. Use the symbols given, not your own!



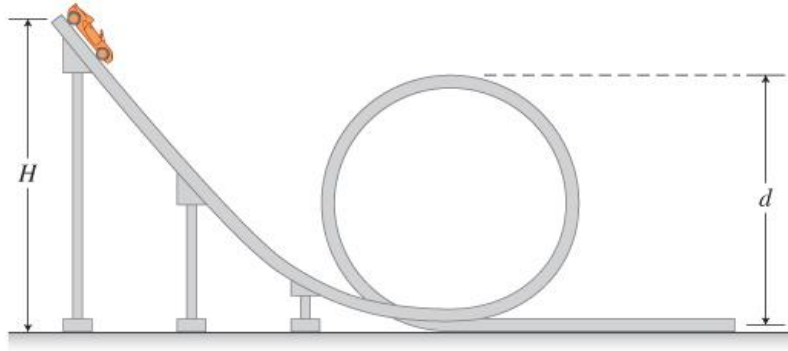
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The car of mass m_c rolls from rest with negligible friction down the curved ramp and around the circular loop. The initial height of the center of mass of the car is at height H . Assume the center of mass of the car is at a height equal to the diameter d of the loop when the car is at the top of the loop.

(a) In terms of the given quantities and any fundamental constants, derive an equation for the speed v_{top} at the top of the loop. (Neglect the rotational kinetic energy of the wheels).



Students, draw anywhere on this slide!



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The car of mass m_c rolls from rest with negligible friction down the curved ramp and around the circular loop. The initial height of the center of mass of the car is at height H . Assume the center of mass of the car is at a height equal to the diameter d of the loop when the car is at the top of the loop.

(a) In terms of the given quantities and any fundamental constants, derive an equation for the speed v_{top} at the top of the loop. (Neglect the rotational kinetic energy of the wheels).

$$K_f + U_{g_f} = U_{g_i} \text{ so apply}$$

$$K = \frac{1}{2}mv^2 \text{ and } \Delta U_g = mg\Delta y$$

$$K_f = \Delta U_g$$

$$\text{and } \frac{1}{2}m_c v_{\text{top}}^2 = m_c g(H - d)$$

$$v_{\text{top}} = \sqrt{2g(H - d)}$$

FRQ Tips and Tricks!--Key Terms!

The words **“sketch”** and **“plot”** relate to graphs:

- **“Sketch”** means to draw a graph that illustrates key trends in a particular relationship, such as slope, curvature, intercept(s), or asymptote(s).
 - Numerical scaling or specific data points are **not** required in a sketch.
- **“Plot”** means to draw the data points given in the problem on the grid provided, either using the given scale or indicating the scale and units when none are provided.

Important Concepts - Quantitative/ Qualitative Translation (QQT)

QUALITATIVE ANALYSIS

- Determine the **physics principles** in the question: Energy? Momentum? Simple harmonic motion? Forces?
- Qualitative does **not** mean writing an equation in words! You have to explain how the physics principles **apply** to the given scenario
- Write using bullet points (**complete thoughts**)

QUANTITATIVE ANALYSIS

- **Derive** all your expressions starting from **physics principles** (conservation laws, net force equations, etc.). Do not take shortcuts!

Important Concepts - (QQT)

TRANSLATION

- Look for **relationships** among variables, are they directly proportional? Inversely proportional?
- Check for a clear **correlation** between your **qualitative** statements and your **quantitative** expressions

GRAPHING TIPS

- Always include: scale, variables, units
- Bring a ruler to draw best-fit lines. Never connect the dots!
- When data is not linear, draw a smooth curve through the data
- If two curves are on the same graph clearly label each curve

GRAPHICAL ANALYSIS

- **Use the graph** to determine information: **slope**, **area** under the curve and/or **intercepts**
- When calculating a slope use a pair of points directly from your **best-fit line**
- If the graph is unfamiliar, look at the **units** of slope and/or area to determine if it represents a **physical quantity**
- If a graph or a set of data is given, look for **outliers** which should be examined to determine if they are bad data points that should be omitted

Important Concepts All Types of FRQ

Most common SLOPES:

- The slope of a **position-time** graph gives the **velocity**
- The slope of a **velocity-time** graph gives the **acceleration**
- The slope of a **force-elongation** graph gives the **spring constant**

Most common AREAS under the curve:

- The area under the curve of a **velocity-time** graph gives the **displacement**
- The area under the curve of an **acceleration-time** graph gives the **change in velocity**
- The area under the curve of a **force-time** graph gives the **impulse** (change in momentum)
- The area under the curve of a **force-distance** graph gives the **work** done

Important Concepts All Types of FRQ

FBDs

- Use a ruler to draw the forces: the **tip of the arrow** shows direction
- Start your arrows on the **center** of the dot given (point particle model)
- Do **NOT** draw components in an FBD
- The forces on extended objects (rotational motion) should be drawn at the **point of application of the force**. Do not use the point particle model!
- **Label** the forces appropriately: F_g , F_N , F_f , F_T , F_S

FBDs and NET FORCE

- **Never** write F_c (**centripetal force**) on an FBD! Write the actual forces that keep the object in circular motion
- If there are two forces exerted in the same direction, draw **TWO** arrows: one for each force!
- Write the **net force** (ΣF) for each coordinate axis
- Always write what the net force is **equal to**: either zero or equal to 'ma', or 'mv²/r'

Paragraph Length Response

✓ **B** – *Basic Physics* or *Basic Principles* or *Basic Equations* – State a principle of physics, law of physics, or physics equation that is relevant to the situation presented. There are many ways to say the same statement of basic physics. A set of equations that apply for an object moving with constant velocity could be any of the following: $v = d/t$, $d = vt$, $t = d/v$, “velocity equals distance over time”, or “(time) = (distance)/(velocity)”. A physical principle that says the same thing might be “time and speed are inversely proportional” or “going faster results in less time of travel”. Stating important definitions also counts as basic principles, such as “if an object has more mass, then it is more difficult to make the object reach a certain speed” or “an impatient person wants things to take less time”.

Paragraph Length Response

✅ **C** – *Cite Evidence*– In addition to making generally true statements of physics, talk about the specifics of the situation presented. Talk about physics quantities like distance, time, velocity, force, acceleration, energy, etc. If you are presented with two cases, two objects, or two different moments in time (like “before” and “after” something happens), then *compare* (say what physics quantities are the same between the two) and *contrast* (say what physics quantities are different, and how they are different; in which case is the quantity more or less than the other?).

✅ **D** – *Draw it together* – Once you set out your basic physics and your specifics of the situation at hand, show logically how the two together connect to your answer.

Paragraph Length Response

✓ **A** – *Answer or Assertion* – Did you answer the question posed or make a clear assertion? On an AP Physics exam free-response, often you will be asked a question and then told to “explain your reasoning” or “justify your answer”. The “Answer” may be marking a space next to one answer out of a possible set of answers. For example, there might be a ball rolling off of a table and the problem says that the speed that the ball leaves the table will be increased. The question might be “How will the time it takes for the ball to reach the floor change?” and the answers suggested would be “increase”, “decrease”, and “remain the same”. Each of those possible answers will have a space next to it and you are to mark one of those spaces. This is your **answer** to the question. Note: Some free-response items give you the answer (or say what will happen in the situation) and simply ask you to explain it. In these cases, there is no “answer” and you’ll only need to do steps B, C, and D.

A student drops a bouncy ball from a 1-meter height and the ball bounces back up to the .8 meter mark. In a clear, coherent paragraph-length response that may also contain figures and/or equations, explain why the ball doesn't return to its original height.

Basic Physics



Students, draw anywhere on this slide!

A student drops a bouncy ball from a 1-meter height and the ball bounces back up to the .8 meter mark. In a clear, coherent paragraph-length response that may also contain figures and/or equations, explain why the ball doesn't return to its original height.

Cite Evidence



Students, draw anywhere on this slide!

A student drops a bouncy ball from a 1-meter height and the ball bounces back up to the .8 meter mark. In a clear, coherent paragraph-length response that may also contain figures and/or equations, explain why the ball doesn't return to its original height.

Draw it together...and answer the question!



Students, draw anywhere on this slide!

Experimental Design Questions!

Use D^2 (DSQuARE)

- **D**igram
- **S**etup
- **Q**uantities
- **A**pparatus
- **R**epeated independent variable
- **E**rror Reduction

The Question

How could you find the spring constant value for a jumping toy using conservation of energy? Provide enough detail so that another student could replicate the experiment, including any steps necessary to reduce experimental uncertainty. As needed, use the symbols defined in the table and/or include a simple diagram of the setup.



How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D^2 (D Square)--D

D – *Diagram With Labels* – Draw a diagram labeling all of the equipment and materials as they would appear before any trials take place (or in the middle of a trial).



Students, draw anywhere on this slide!

How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D^2 (D Square)--Diagram

D – *Diagram With Labels* – Draw a diagram labeling all of the equipment and materials as they would appear before any trials take place (or in the middle of a trial).

 motion detector



How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D^2 (D Square)--SQuA

S – *Set-up* – Briefly explain anything about the experimental set-up that your diagram doesn't clearly convey. (There is no reason to say anything in words that can be said by your diagram.) Then say what happens to the set-up (Example: "the block is released from rest and slides down the incline" or something like that).

Qu – *Quantities* – State what quantities are to be directly measured.

A – *Apparatus* – State what apparatus (equipment) is used to measure each directly-measured quantity.

(Example: "Use a meterstick to measure the height of the top of the ramp. Use a stopwatch to measure the time it takes for the block to go from the top of the ramp to the bottom.")

How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D² (D Square)--SQuA

$$k = \frac{mv^2}{x^2}$$

$$k = \frac{2mg\Delta y}{x^2}$$



Students, draw anywhere on this slide!

How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D² (D Square)--SQuA

$$k = \frac{mv^2}{x^2}$$

1. Measure the mass of the toy with the balance.
2. Compress the toy on the table and measure the compression with the meter stick.
3. Let go of the toy and when it rises record the velocity of the toy using the motion detector.

$$k = \frac{2mg\Delta y}{x^2}$$

1. Measure the mass of the toy with the balance.
2. Compress the toy on the table and measure the compression with the meter stick.
3. Let go of the toy and when it rises record the maximum height of the toy using the meter stick (or motion detector). To be more precise, use a cell phone slow motion camera to better see the height.

How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D² (D Square)--R, E

R – *Repeated Independent Variable* – State how trials are to be repeated, including what quantity will change (and how) and what will not change.

(Example: “For five trials, increase the height of the top of the ramp on each trial but keep the length of the ramp the same.”)

E – *Error Reduction* – State how experimental error can be reduced.

(Example: Do each run three times and average.)

How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D² (D Square)--R, E

$$k = \frac{mv^2}{x^2}$$

$$k = \frac{2mg\Delta y}{x^2}$$



Students, draw anywhere on this slide!

How could you find the spring constant value for a jumping toy using conservation of energy?

Experimental Procedure = D² (D Square)--R, E

$$k = \frac{mv^2}{x^2}$$

4. Repeat the steps but increase the compression each time until 7 data points are obtained.
5. For each run, do three trials and average the data.

$$k = \frac{2mg\Delta y}{x^2}$$

4. Repeat the steps but increase the compression each time until 7 data points are obtained.
5. For each run, do three trials and average the data.

Sample 1

Ben D. Straw and Ann A. Bell are trying to determine the acceleration of a toy car. Assuming they have equipment commonly used in a Physics lab, explain a procedure in which they might find the acceleration of the toy car.

D

S-Qu-A



Students, draw anywhere on this slide!

Sample 1

Ben D. Straw and Ann A. Bell are trying to determine the acceleration of a toy car. Assuming they have equipment commonly used in a Physics lab, explain a procedure in which they might find the acceleration of the toy car.

R

E



Students, draw anywhere on this slide!

Sample 1

Ben D. Straw and Ann A. Bell are trying to determine the acceleration of a toy car. How will they use the data to determine the acceleration of the toy car?

Data Table (You do NOT need to use all of the boxes!)



Students, draw anywhere on this slide!

Sample 1

Ben D. Straw and Ann A. Bell are trying to determine the acceleration of a toy car. How will they use the data to determine the acceleration of the toy car?

Graph



Students, draw anywhere on this slide!

Sample 1

Ben D. Straw and Ann A. Bell are trying to determine the acceleration of a toy car.
How will they use the data to determine the acceleration of the toy car?

Acceleration Determination



Students, draw anywhere on this slide!



The logo for fiveable, featuring the word "fiveable" in a bold, sans-serif font. "five" is white and "able" is dark blue. The text is contained within a light blue speech bubble shape that points downwards.

fiveable

AP Physics 1

Finale Part 5--MCQ Tips and Tricks with GimKit
Challenge!

- **5:00-5:55 Energy!**
- **6:00-6:55 Momentum!**
- **7:00-7:55 Simple Harmonic Motion**
- **8:00-8:55 FRQ Tips and Tricks**
- **9:00-9:00 MCQ Tips and Tricks
with GimKit Challenge!**

Multiple Choice Question Tips and Tricks!

Use a REGULAR PENCIL (don't use mechanical!)

You DO get to use a calculator!

You DO get to use the Equation Sheet!

There are 50 MCQ on the exam.

- 45 are single response (one answer)
- 5 are multiple response (two answers)
- You have 90 minutes to complete the MCQ
 - That's 1.8 minutes PER QUESTION (1 minute, 48 seconds)
 - Don't rush BUT make sure you watch the clock!

Multiple Choice Question Tips and Tricks!

WRITE ON THE EXAM!

- Do the questions you feel most comfortable with FIRST (don't go in order!)
- Mark off answers that you can eliminate (look for the wrong unit, wrong direction, etc.)
- Circle questions you want to go "back to"
- DRAW A PICTURE of the situation--that can sometimes help!
- If the question requires a lot of "math" (Algebra manipulation), leave it for later (go back to it)

Multiple Choice Question Tips and Tricks!

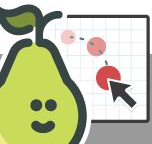
- Read each question carefully. **Circle key words**
- Think of **physics principles** before using equations: is this an energy scenario? is the object moving in a circle? etc.
- If the answer is **too obvious** (*Mickey Mouse type*), read the question again, the question might **not** be trivial!
- If you do not know the answer, try **eliminating** incorrect answers
- Use **proportional reasoning** if possible rather than plug & chug
- Watch out for the MULTIPLE CORRECT section and be sure to select **two** answers
- Do **NOT** leave any question blank!!
 - There is NO penalty for guessing! So make an **educated** guess!

Gimkit Battle!



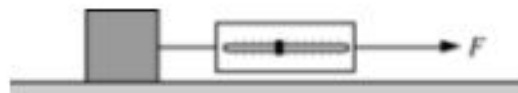
In a lab, a block weighing 80 N is attached to a spring scale, and both are pulled to the right on a horizontal surface, as shown above. Friction between the block and the surface is negligible. What is the acceleration of the block when the scale reads 32 N?

- A** 2.0 m/s^2
- B** 2.5 m/s^2
- C** 4.0 m/s^2
- D** 6.0 m/s^2



Students, drag the icon!





In a lab, a block weighing 80 N is attached to a spring scale, and both are pulled to the right on a horizontal surface, as shown above. Friction between the block and the surface is negligible. What is the acceleration of the block when the scale reads 32 N?

A 2.0 m/s^2

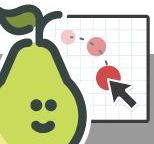
B 2.5 m/s^2

 **C** 4.0 m/s^2

D 6.0 m/s^2

An object attached to one end of a string moves in a circle at constant speed. Which of the following is correct?

- A** The object is accelerating as it moves.
- B** The object's velocity is the same as its speed.
- C** The object does not require a force to keep its state of circular motion.
- D** If the string breaks, the object will keep its circular motion.



An object attached to one end of a string moves in a circle at constant speed. Which of the following is correct?



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B

The object's velocity is the same as its speed.

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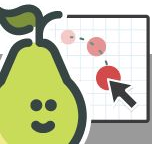
The object does not require a force to keep its state of circular motion.

D

If the string breaks, the object will keep its circular motion.

How does an air mattress protect a stunt person landing on the ground after a stunt?

- A. It reduces the momentum change of the stunt person.
- B. It increases the momentum change of the stunt person.
- C. It shortens the stopping time of the stunt person and increases the force applied during the landing.
- D. It lengthens the stopping time of the stunt person and reduces the force applied during the landing.



Students, drag the icon!



How does an air mattress protect a stunt person landing on the ground after a stunt?

- A. It reduces the momentum change of the stunt person.
- B. It increases the momentum change of the stunt person.
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-  D. It lengthens the stopping time of the stunt person and reduces the force applied during the landing.

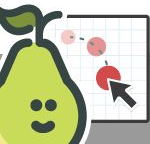
A force of constant magnitude F and fixed direction acts on an object of mass m that is initially at rest. If the force acts for a time interval Δt over a displacement Δx , what is the magnitude of the resultant change in the linear momentum of the object?

A $F \Delta t$

B $F \Delta x$

C $F \Delta t / m$

D $F \Delta x / m$



Students, drag the icon!



A force of constant magnitude F and fixed direction acts on an object of mass m that is initially at rest. If the force acts for a time interval Δt over a displacement Δx , what is the magnitude of the resultant change in the linear momentum of the object?



A

$$F \Delta t$$

B

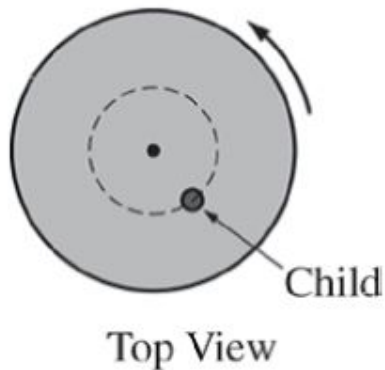
$$F \Delta x$$

C

$$F \Delta t / m$$

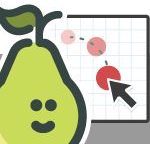
D

$$F \Delta x / m$$



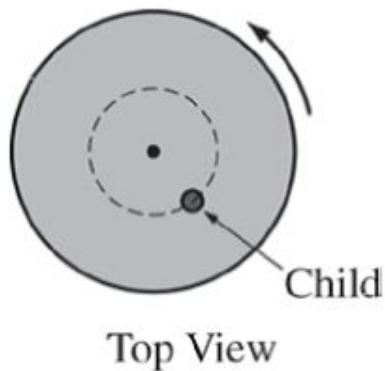
The diagram above shows a top view of a child of mass M on a circular platform of mass $2M$ that is rotating counterclockwise. Assume the platform rotates without friction. Which of the following describes an action by the child that will increase the angular speed of the platform-child system and gives the correct reason why?

- ☐ A The child moves toward the center of the platform, increasing the total angular momentum of the system.
- ☐ B The child moves toward the center of the platform, decreasing the rotational inertia of the system.
- ☐ C The child moves away from the center of the platform, increasing the total angular momentum of the system.
- ☐ D The child moves away from the center of the platform, decreasing the rotational inertia of the system.



Students, drag the icon!





The diagram above shows a top view of a child of mass M on a circular platform of mass $2M$ that is rotating counterclockwise. Assume the platform rotates without friction. Which of the following describes an action by the child that will increase the angular speed of the platform-child system and gives the correct reason why?

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- ☐ C The child moves away from the center of the platform, increasing the total angular momentum of the system.
- ☐ D The child moves away from the center of the platform, decreasing the rotational inertia of the system.



A railroad car of mass m is moving with speed u when it collides with and connects to a second railroad car of mass $3m$, initially at rest, as shown above. How do the speed and kinetic energy of the connected cars compare to those of the single car of mass m before the collision?

A

Speed	Kinetic Energy
Less	Less

B

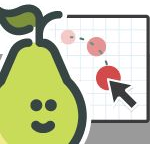
Speed	Kinetic Energy
Less	The same

C

Speed	Kinetic Energy
The same	Less

D

Speed	Kinetic Energy
The same	The same



Students, drag the icon!





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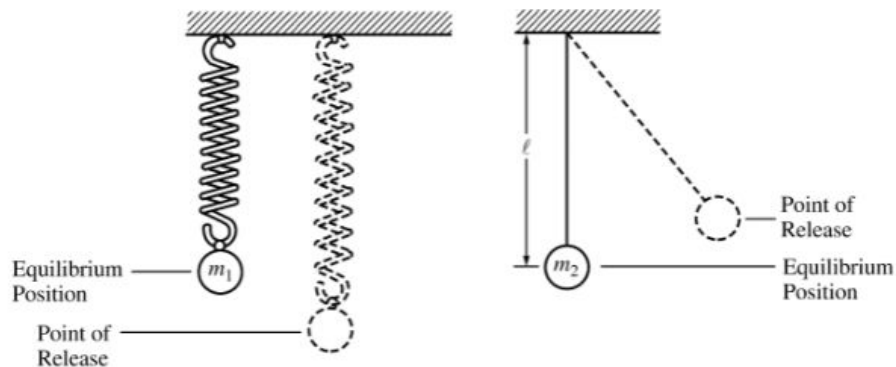
Speed	Kinetic Energy
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D

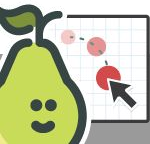
Speed	Kinetic Energy
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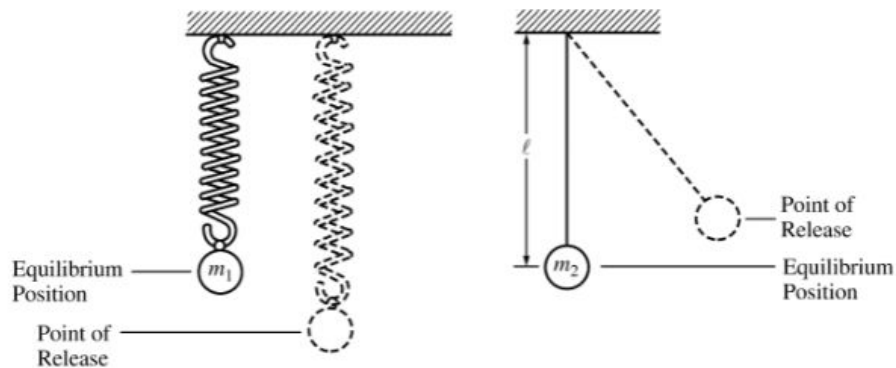
A sphere of mass m_1 , which is attached to a spring, is displaced downward from its equilibrium position as shown above left and released from rest. A sphere of mass m_2 , which is suspended from a string of length l is displaced to the right as shown above right and released from rest so that it swings as a simple pendulum with small amplitude. Assume that both spheres undergo simple harmonic motion.

Which of the following is true for both spheres?

- ☐ A The maximum kinetic energy is attained as the sphere passes through its equilibrium position.
- ☐ B The maximum kinetic energy is attained as the sphere reaches its point of release.
- ☐ C The minimum gravitational potential energy is attained as the sphere passes through its equilibrium position.
- ☐ D The maximum gravitational potential energy is attained when the sphere reaches its point of release.



Students, drag the icon!

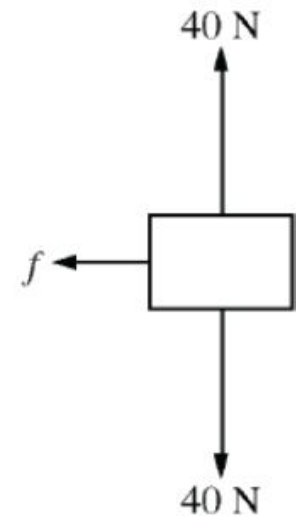


A sphere of mass m_1 , which is attached to a spring, is displaced downward from its equilibrium position as shown above left and released from rest. A sphere of mass m_2 , which is suspended from a string of length l is displaced to the right as shown above right and released from rest so that it swings as a simple pendulum with small amplitude. Assume that both spheres undergo simple harmonic motion.

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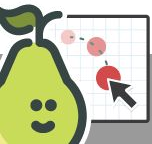


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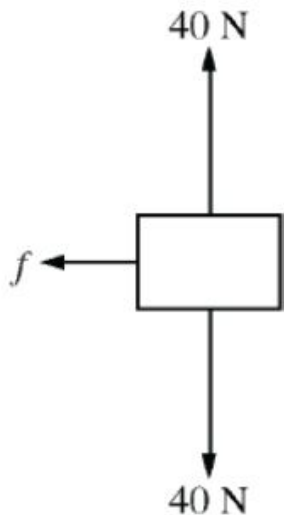
The figure above shows the forces exerted on a block that is sliding on a horizontal surface: the gravitational force of 40 N, the 40 N normal force exerted by the surface, and a frictional force exerted to the left. The coefficient of friction between the block and the surface is 0.20. The acceleration of the block is most nearly

- ☐ A 1.0 m/s^2 to the right
- ☐ B 1.0 m/s^2 to the left
- ☐ C 2.0 m/s^2 to the right
- ☐ D 2.0 m/s^2 to the left



Students, drag the icon!





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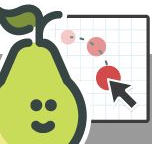
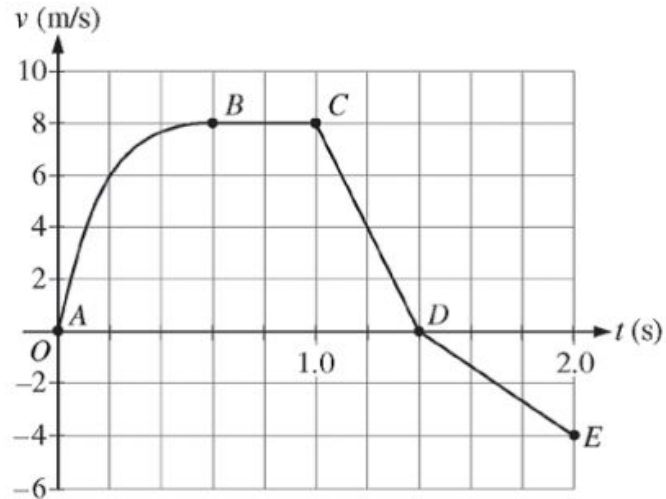
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- ☐ C 2.0 m/s^2 to the right
- ☒ D 2.0 m/s^2 to the left



A cart is constrained to move along a straight line. A varying net force along the direction of motion is exerted on the cart. The cart's velocity v as a function of time t is shown in the graph above. The five labeled points divide the graph into four sections.

During some part of the motion, the work done on the cart is negative. What feature of the motion indicates this?

- A** The speed is increasing.
- B** The speed is decreasing.
- C** The acceleration is positive.
- D** The acceleration is negative.



Students, drag the icon!



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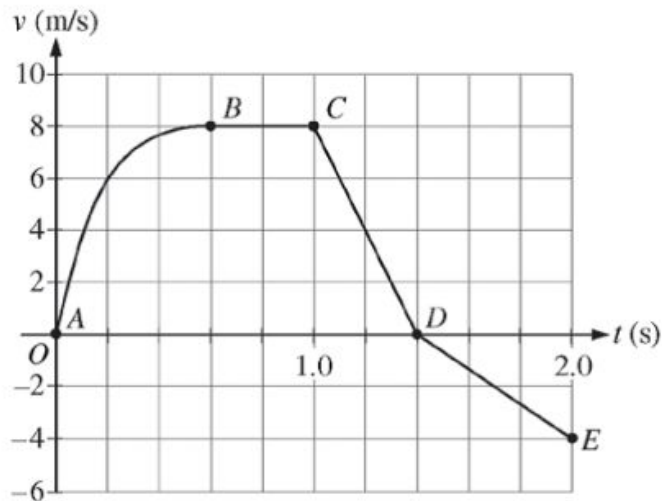
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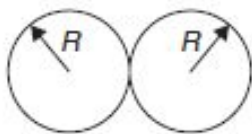
C

The acceleration is positive.

D

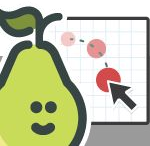
The acceleration is negative.





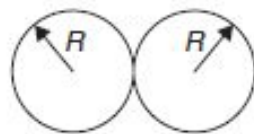
Two solid spheres of radius R made of the same type of steel are placed in contact, as shown in the figures above. The magnitude of the gravitational force that they exert on each other is F_1 . When two other solid spheres of radius $3R$ made of this steel are placed in contact, what is the magnitude of the gravitational force that they exert on each other?

- A** F_1
- B** $3F_1$
- C** $9F_1$
- D** $81F_1$



Students, drag the icon!





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A F_1

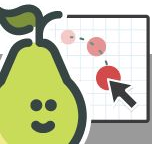
B $3F_1$

C $9F_1$

 **D** $81F_1$

While traveling in its elliptical orbit around the Sun, Mars gains speed during the part of the orbit where it is getting closer to the Sun. Which of the following can be used to explain this gain in speed?

- A** As Mars gets closer to the Sun, the Mars–Sun system loses potential energy and Mars gains kinetic energy.
- B** A component of the gravitational force exerted on Mars is perpendicular to the direction of motion, causing an acceleration and hence a gain in speed along that direction.
- C** The torque exerted on Mars by the Sun during this segment of the orbit increases the Mars–Sun system’s angular momentum.
- D** The centripetal force exerted on Mars is greater than the gravitational force during this segment of the orbit, causing Mars to gain speed as it gets closer to the Sun.



Students, drag the icon!



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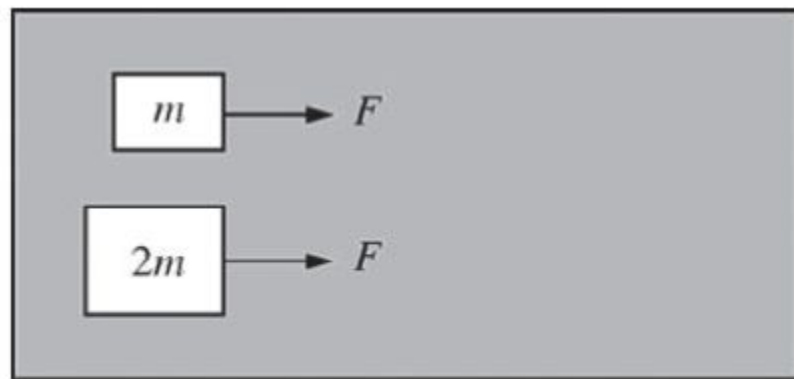
The torque exerted on Mars by the Sun during this segment of the orbit increases the Mars–Sun system's angular momentum.

D

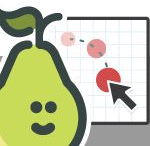
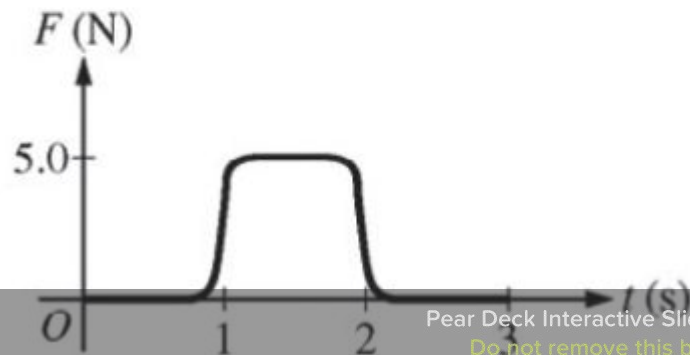
The centripetal force exerted on Mars is greater than the gravitational force during this segment of the orbit, causing Mars to gain speed as it gets closer to the Sun.

Two blocks, of mass m and $2m$, are initially at rest on a horizontal frictionless surface. A force F is exerted individually on each block, as shown above. The graph shows how F varies with time t . Which block has the greatest average power provided to it between $t = 0$ s and $t = 3$ s?

- A** The block of mass m
- B** The block of mass $2m$
- C** Both blocks have the same power provided to them.
- D** It cannot be determined without knowing the ratio of the maximum force to the mass m .



Top View



Two blocks, of mass m and $2m$, are initially at rest on a horizontal frictionless surface. A force F is exerted individually on each block, as shown above. The graph shows how F varies with time t . Which block has the greatest average power provided to it between $t = 0$ s and $t = 3$ s?



A

The block of mass m

B

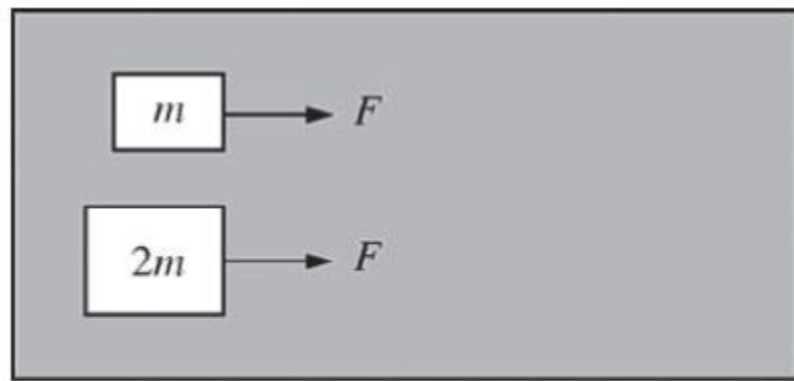
The block of mass $2m$

C

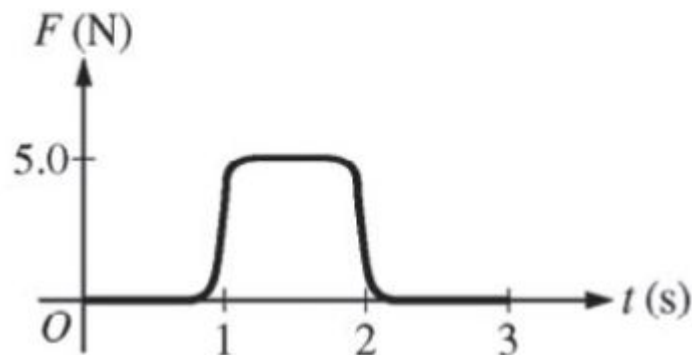
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D

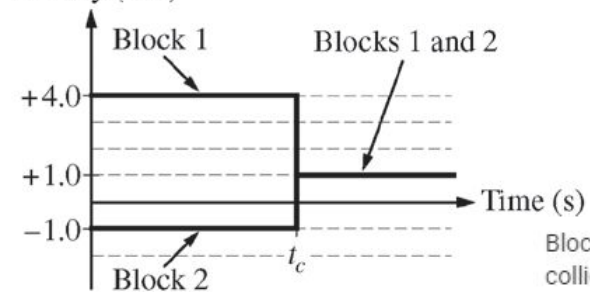
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Top View



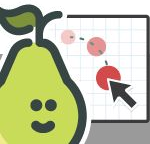
Velocity (m/s)



Block 1 of mass m_1 and block 2 of mass m_2 are sliding along the same line on a horizontal frictionless surface when they collide at time t_c . The graph above shows the velocities of the blocks as a function of time.

How does the kinetic energy of the two-block system after the collision compare with its kinetic energy before the collision, and why?

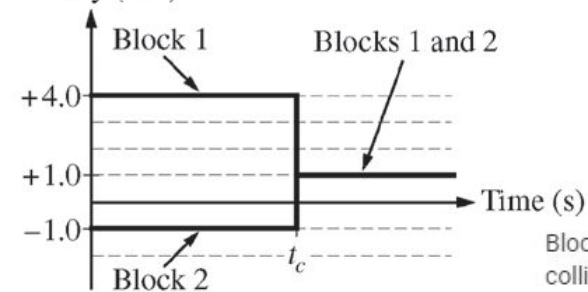
- ☐ A It is less, because the blocks have the same velocity after the collision, so some of their kinetic energy was transformed into internal energy.
- ☐ B It is less, because the blocks have velocities in opposite directions before the collision, so some of their kinetic energy cancels.
- ☐ C It is the same, because the collision was instantaneous, so the effect of external forces during the collision is negligible.
- ☐ D It is the same, because the blocks have the same velocity after the collision, and there is no friction acting on them.



Students, drag the icon!



Velocity (m/s)



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How does the kinetic energy of the two-block system after the collision compare with its kinetic energy before the collision, and why?



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It is less, because the blocks have the same velocity after the collision, so some of their kinetic energy was transformed into internal energy.

B

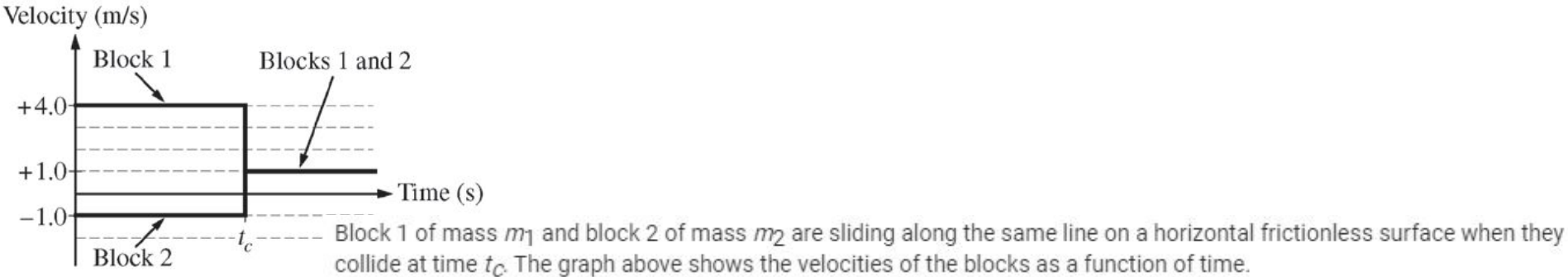
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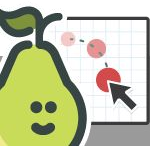
D

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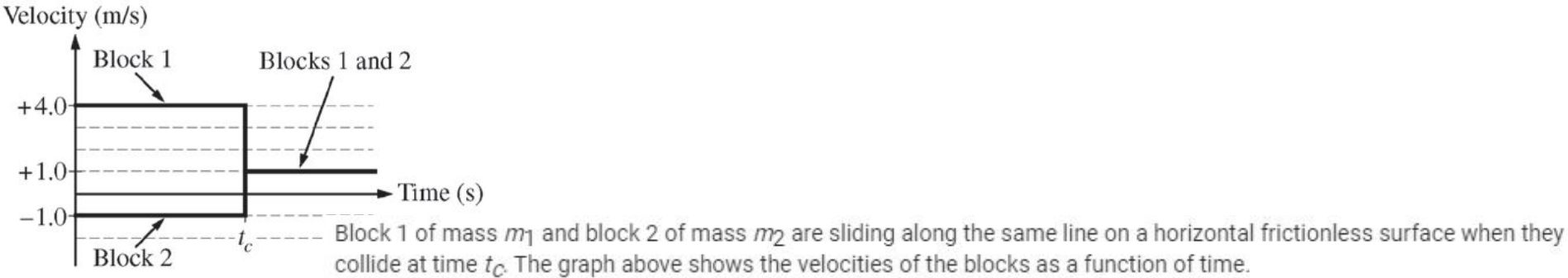
Which block has the greater mass, and what information indicates this?

- ☐ A Block 1, because it had a greater speed before the collision.
- ☐ B Block 1, because the velocity after the collision is in the same direction as its velocity before the collision.
- ☐ C Block 2, because it had a smaller speed before the collision.
- ☐ D Block 2, because the final velocity is closer to the initial velocity of block 2 than it is to the initial velocity of block 1.



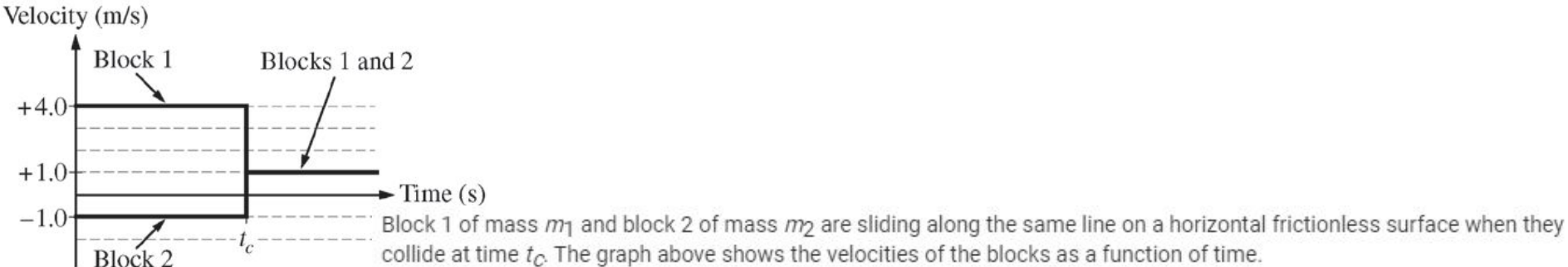
Students, drag the icon!





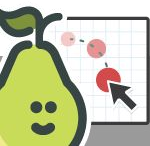
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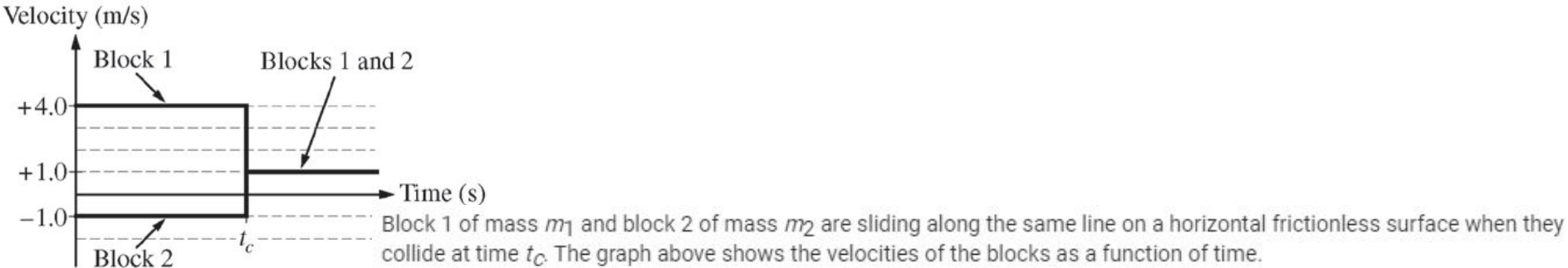
Which of the following is true of the motion of the center of mass of the two-block system during the time shown?

- ☐ A The center of mass does not move because the blocks are moving in opposite directions before the collision.
- ☐ B The center of mass moves at a constant velocity of +1.0 m/s because there is no friction acting on the system.
- ☐ C The center-of-mass velocity starts out greater than +1.0 m/s but decreases to +1.0 m/s during the collision because the collision is inelastic.
- ☐ D The center-of-mass velocity increases as the blocks get closer together, and then becomes constant after the collision.



Students, drag the icon!





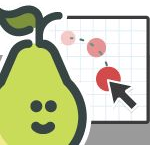
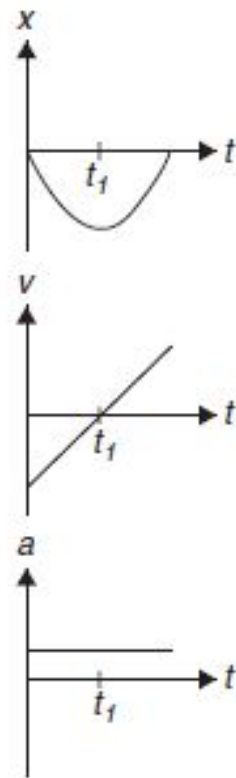
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- D** The center-of-mass velocity increases as the blocks get closer together, and then becomes constant after the collision.

The graphs above represent the position x , velocity v , and acceleration a as a function of time t for a marble moving in one dimension. Which of the following could describe the motion of the marble?

- A** Rolling along the floor and then bouncing off a wall
- B** Rolling down one side of a bowl and then rolling up the other side
- C** Rolling up a ramp and then rolling back down
- D** Falling and then bouncing elastically off a hard floor



Students, drag the icon!



The graphs above represent the position x , velocity v , and acceleration a as a function of time t for a marble moving in one dimension. Which of the following could describe the motion of the marble?



Rolling along the floor and then bouncing off a wall



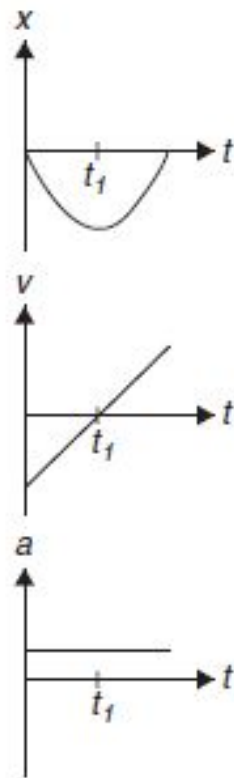
Rolling down one side of a bowl and then rolling up the other side



Rolling up a ramp and then rolling back down

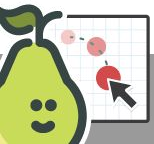
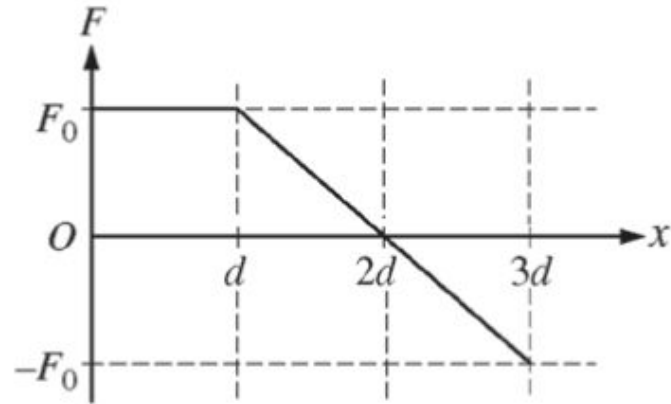


Falling and then bouncing elastically off a hard floor



An object is moving in the positive x -direction while a net force directed along the x -axis is exerted on the object. The figure above shows the force as a function of position. What is the net work done on the object over the distance shown?

- A** $F_0 d$
- B** $3F_0 d/2$
- C** $2F_0 d$
- D** $4F_0 d$



Students, drag the icon!



An object is moving in the positive x -direction while a net force directed along the x -axis is exerted on the object. The figure above shows the force as a function of position. What is the net work done on the object over the distance shown?



A

$F_0 d$

B

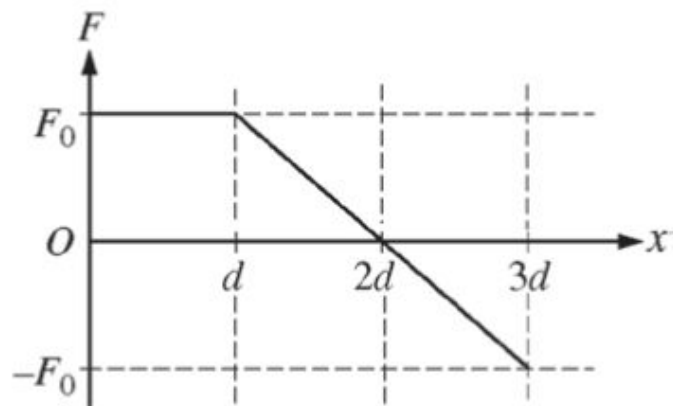
$3F_0 d/2$

C

$2F_0 d$

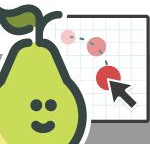
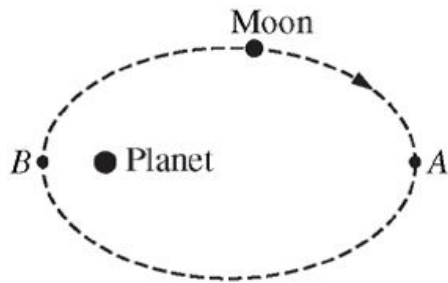
D

$4F_0 d$



A moon is in an elliptical orbit about a planet as shown above. At point A the moon has speed μ_A and is at distance R_A from the planet. At point B the moon has speed μ_B . Which of the following explains a correct method for determining the distance of the moon from the planet at point B in terms of the given quantities?

- A** Conservation of angular momentum, because the gravitational force exerted by the moon on the planet is the same as that exerted by the planet on the moon
- B** Conservation of angular momentum, because the gravitational force exerted on the moon is always directed toward the planet
- C** Conservation of energy, because the gravitational force exerted on the moon is always directed toward the planet
- D** Conservation of energy, because the gravitational force exerted by the moon on the planet is the same as that exerted by the planet on the moon



Students, drag the icon!



A moon is in an elliptical orbit about a planet as shown above. At point A the moon has speed μ_A and is at distance R_A from the planet. At point B the moon has speed μ_B . Which of the following explains a correct method for determining the distance of the moon from the planet at point B in terms of the given quantities?



Conservation of angular momentum, because the gravitational force exerted by the moon on the planet is the same as that exerted by the planet on the moon



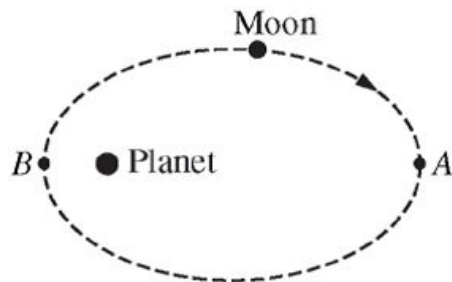
Conservation of angular momentum, because the gravitational force exerted on the moon is always directed toward the planet



Conservation of energy, because the gravitational force exerted on the moon is always directed toward the planet



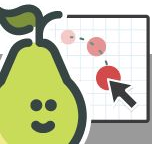
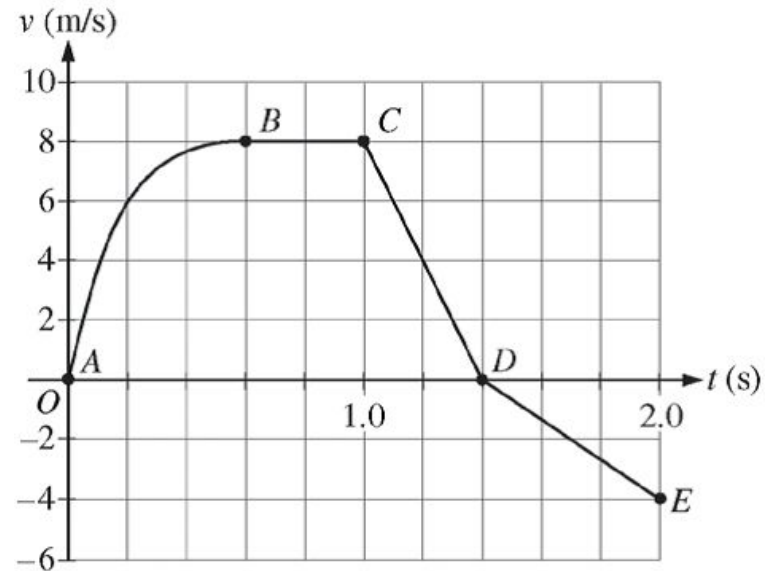
Conservation of energy, because the gravitational force exerted by the moon on the planet is the same as that exerted by the planet on the moon



A cart is constrained to move along a straight line. A varying net force along the direction of motion is exerted on the cart. The cart's velocity v as a function of time t is shown in the graph above. The five labeled points divide the graph into four sections.

For which segment does the cart move the greatest distance?

- ☐ **A** AB
- ☐ **B** BC
- ☐ **C** CD
- ☐ **D** DE



Students, drag the icon!

A cart is constrained to move along a straight line. A varying net force along the direction of motion is exerted on the cart. The cart's velocity v as a function of time t is shown in the graph above. The five labeled points divide the graph into four sections.

For which segment does the cart move the greatest distance?



A

AB

B

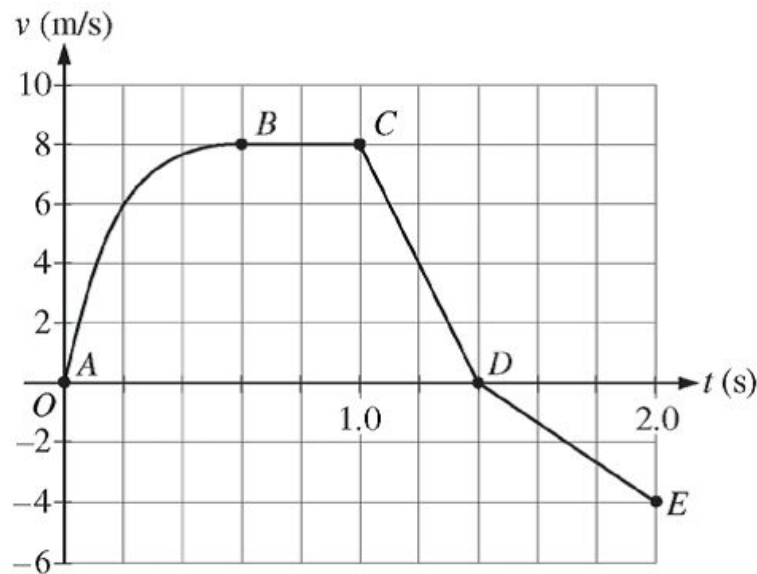
BC

C

CD

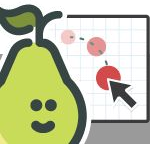
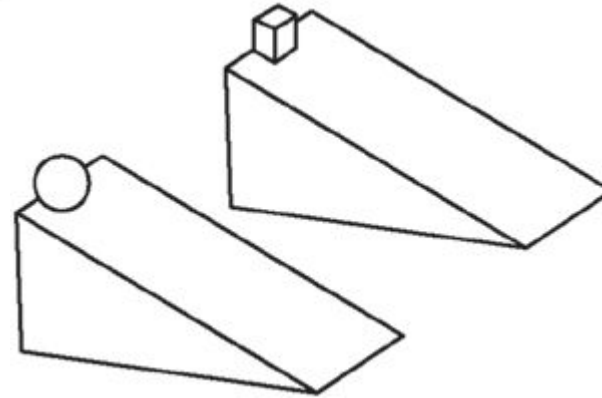
D

DE



Two objects are released from rest at the top of ramps with the same dimensions, as shown in the figure above. The sphere rolls down one ramp without slipping. The small block slides down the other ramp without friction. Which object reaches the bottom of its ramp first, and why?

- A** The sphere, because it gains rotational kinetic energy, but the block does not
- B** The sphere, because it gains mechanical energy due to the torque exerted on it, but the block does not
- C** The block, because it does not lose mechanical energy due to friction, but the sphere does
- D** The block, because it does not gain rotational kinetic energy, but the sphere does



Students, drag the icon!



Two objects are released from rest at the top of ramps with the same dimensions, as shown in the figure above. The sphere rolls down one ramp without slipping. The small block slides down the other ramp without friction. Which object reaches the bottom of its ramp first, and why?

A

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B

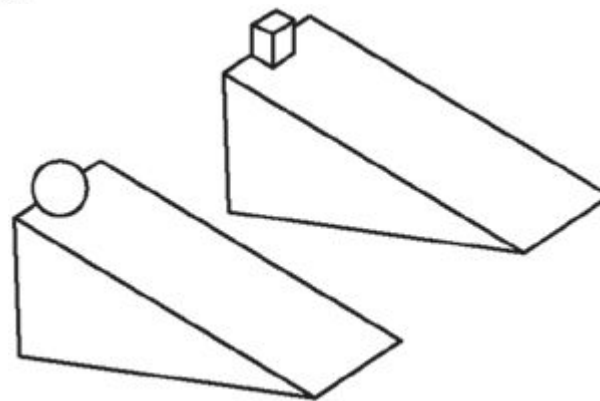
The sphere, because it gains mechanical energy due to the torque exerted on it, but the block does not

C

The block, because it does not lose mechanical energy due to friction, but the sphere does

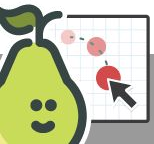
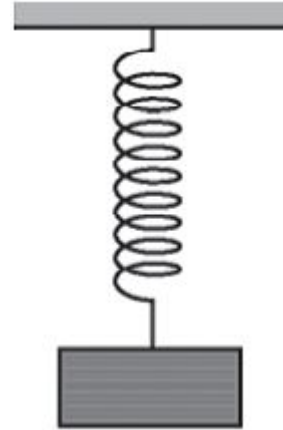
D

The block, because it does not gain rotational kinetic energy, but the sphere does



A block of known mass hanging from an ideal spring of known spring constant is oscillating vertically. A motion detector records the position, velocity, and acceleration of the block as a function of time. Which of the following indicates the measured quantities that are sufficient to determine whether the net force exerted on the block equals the vector sum of the individual forces?

- A** Acceleration only
- B** Acceleration and position only
- C** Acceleration and velocity only
- D** Acceleration, position, and velocity

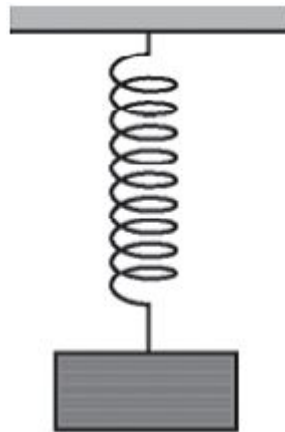


Students, drag the icon!



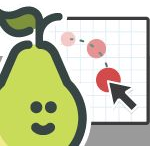
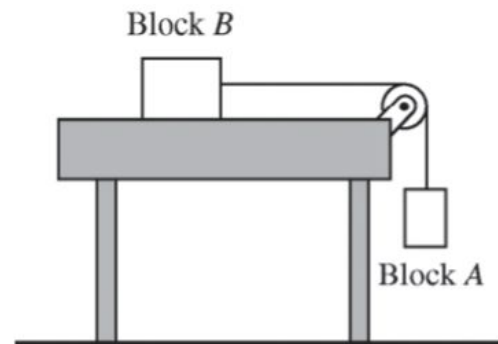
A block of known mass hanging from an ideal spring of known spring constant is oscillating vertically. A motion detector records the position, velocity, and acceleration of the block as a function of time. Which of the following indicates the measured quantities that are sufficient to determine whether the net force exerted on the block equals the vector sum of the individual forces?

- ☐ A Acceleration only
- ☒ B Acceleration and position only
- ☐ C Acceleration and velocity only
- ☐ D Acceleration, position, and velocity



Block *A* hangs from a light string that passes over a light pulley and is attached to block *B*, which is on a level horizontal frictionless table, as shown above. Students are to determine the mass of block *B* from the motion of the two-block system after it is released from rest. They plan to measure the time block *A* takes to reach the floor. The students must also take which of the following measurements to determine the mass of block *B*?

- A** Only the mass of block *A*
- B** Only the distance block *A* falls to reach the floor
- C** Only the mass of block *A* and the distance block *A* falls to reach the floor
- D** The mass of block *A*, the distance block *A* falls to reach the floor, and the radius of the pulley



Students, drag the icon!



Block *A* hangs from a light string that passes over a light pulley and is attached to block *B*, which is on a level horizontal frictionless table, as shown above. Students are to determine the mass of block *B* from the motion of the two-block system after it is released from rest. They plan to measure the time block *A* takes to reach the floor. The students must also take which of the following measurements to determine the mass of block *B*?



Only the mass of block *A*



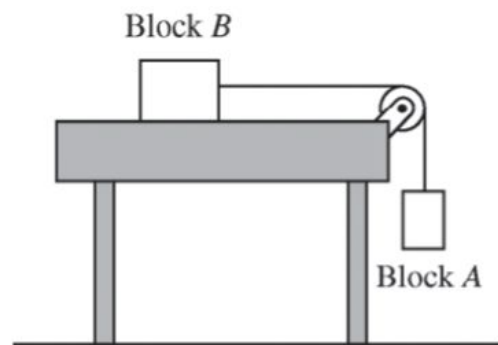
Only the distance block *A* falls to reach the floor



Only the mass of block *A* and the distance block *A* falls to reach the floor



The mass of block *A*, the distance block *A* falls to reach the floor, and the radius of the pulley



Keep the conversation going in Discord!

<http://5able.me/fiveable-forums>

